Causal Evolution Dynamics: Formalizing the Boundaries of Grounding and Hysteresis in Complex Systems

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Abstract

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Draft 1.0 — abstract. Every number quoted here is produced by `evolution_sim.py` *and re-checked by* `--selftest` *(10 invariant groups) and* `--audit-doc` *(28 claims). No number is typed from memory.*

Abstract

Complex systems increasingly report on their own success. When the report and the delivery come from different places, a system can keep recording completion after the work has stopped landing. We give this failure a formal treatment rather than a rhetorical one.

We build a minimal formal system: nine axioms, twelve definitions, ten inference rules, an explicit domain for every variable, and a four-valued verdict (`PASS` / `FAIL` / `INVALID` / `NOT_RUN`) that is a statement about the *scope* of a claim rather than about its success. Two dynamical laws are kept separate throughout, because they are not the same equation: a saturating metabolic law (M1, used while dormant) and a linear accounting law (M2, used while active). Earlier versions of this work treated their concatenation as one "unified equation"; we show they have different fixed points, different stability and different time constants, and that the concatenation is an assumption (A7) rather than a derivation.

Three results survive the audit.

1. **The deadlock boundary is hyperbolic and exact.** Under a quasi-static substrate assumption and a wake rule that requires κ consecutive qualifying cycles, the dormant state is permanent if and only if $\chi < \chi_{\min} = 1/(1-\theta_{\text{wake}})$, where $\chi(v) = Y\phi_v v/(m_0\phi_{\text{dorm}})$ is the charging margin of the agent itself; equivalently $\chi_0 < 1/(f(1-\theta_{\text{wake}}))$ with $\chi_0 = \chi(V_{\max})$ and $f = v/V_{\max}$. The two forms differ by the factor f and must not be quoted interchangeably. Because the uptake law is unimodal, the margin has a finite maximum: $\sup_S \chi(S) = 27.78$ and $\sup_S A_\infty = 0.964$, attained at $S = \sqrt{K_s K_i}$. Deadlock above 0.964 is therefore immune to any amount of substrate — the pinned agent of §2.4 at $\theta = 0.99$ is pinned for that reason — while below it a flood buys at most the factor $v_{\max}/v(S_0) = 1.56$ and rescues only a margin still above $0.64\,\chi_{\min}$. The ceiling comes from the S^2/K_i term, not from a decaying marginal return; we no longer claim the stronger statement, which is false in the range this paper's sweep covers.

2. **A closed system has no hysteresis loop.** With a Lyapunov function $V = S$, a batch system without feed drives $S \to 0$ and then A below any threshold in finite time; the downward edge is autonomous and the upward edge can never fire. The general claim that *any* hysteresis loop needs an external upward trigger is false (van der Pol is a counterexample), and we withdraw it.
3. **A drift detector with power one, and its limits.** Any affine-invariant statistic has power exactly α against linear drift; sliding-window variance has an expectation that does not grow with n. The cumulative potential $\Phi_n = n\Delta_{\text{drift}} + \varepsilon_0 - \varepsilon_n$ has signal-to-noise $n\Delta_{\text{drift}}/(\sqrt{2}\sigma) \to \infty$. It is unbiased and consistent, but **not sufficient** and not efficient: the OLS slope has variance $12\sigma^2/[n(n^2-1)]$, an order of magnitude smaller.

We then audit our own framework and publish the audit. Twenty-eight implicit assumptions are identified; **eleven are load-bearing** — if any one fails, at least one of the results above or one headline number collapses. They are listed in Table 1 with the specific theorem each supports, and the seventeen that are not load-bearing are inventoried with their failure modes. We also withdraw two strong claims of the earlier draft (that multiplicative availability is *necessary* in systems, and that the observed cliff is a *phase transition*), and we downgrade the latter to a conjecture that is, in the independent-agent limit, false.

Finally we report a controlled ablation in a 100-agent sandbox. Type-A commits a fixed scope and reads only its own log; Type-B commits a scope that is a function of the world's verdict. Both face the same capacity trajectory. Type-A stops delivering permanently at cycle 100, dies at 107, and its own log ends 15.56% above what landed. Type-B is stranded at 279. The lifetime ratio is **2.61**, not the 9.19 that the previous draft reported; the difference is a unit error in the wake gate, which we isolate as a three-way sensitivity table (2.61 / 2.18 / 9.19). Wiring the `INVALID` channel into Type-A — and changing nothing else — is sufficient for it to survive the same 300 cycles with a zero log gap. The channel, not the drift, is what the cliff is made of.

The extinction is scheduled rather than emergent: Corollary 4 places each death four to seven cycles after the onset, from the upkeep rates alone.

The algebraic skeleton — the threshold equivalence of Lemma 4, the mutual exclusion of the two modes, and the totality of the four-valued verdict — is machine-checked in Lean 4 (`formal_verification/Core_Theorems.lean`): 0 errors, 0 warnings, no `sorry`, with the `#print axioms` receipts printed from the source rather than summarised here.

We make no claim of universality. The contribution is a boundary that can be checked, three proofs that can be refuted, and a list of the assumptions that would have to hold for either.

Keywords: grounding; hysteresis; deadlock; open systems; self-report;

1. Introduction

Draft 1.0. Section numbers here match `02_Method.md`, `03_Results.md` and `04_Discussion.md`; cross-references have been checked against them.

1.1 The failure mode

A system that reports its own completion has two descriptions of the same cycle: what it says it did, and what landed. When both descriptions are produced by the same process they agree by construction, and agreement is not evidence. The interesting case is when they can diverge — when a commitment is fixed before the capacity is known, and the record is written by the party that made the commitment.

This is not a metaphorical worry. It is the ordinary condition of any system that advertises a scope, is paid for the scope, and observes only its own intent. Three properties make it dangerous rather than merely inaccurate.

The failure is silent. Nothing in the agent's own record changes at the moment delivery stops. A variance alarm on its own series fires zero times, because the series is smooth: it records the claim, and the claim is a constant. The drift is real, monotone, and invisible to every affine-invariant statistic (§2.8).

The failure is late. The delivery threshold and the break-even threshold are different numbers. An agent stops landing work when capacity $<$ claim; it starts losing charge at the slower rate capacity $<$ claim$/(1+\epsilon_m)$. In our shipped scenario those are 0.930 and 0.809 of day-1 capacity, and the agent crosses the first permanently at cycle 100 and the second only after its income has already gone to zero. The distance between the two thresholds is the length of the drought it cannot see.

The failure is absorbing, and the absorption has a ceiling. Once charge falls below the wake threshold, a saturating recharge law still has a positive limit, so "no matter how much substrate" cannot be argued from saturation alone. What closes the recovery is the shape of the uptake law. (Ax1) is unimodal, with its maximum at $S = \sqrt{K_s K_i} = 2$, so the charging margin is bounded: $\sup_S \chi(S) = 27.78$ and $\sup_S A_\infty = 0.964$. Above that ceiling no substrate level restores the agent — this is the regime of the $\theta = 0.99$ example in §2.4, where an agent is pinned with substrate still available. Below the ceiling, flooding buys a bounded factor and no more, and it rescues only a margin that has not yet fallen below $\chi_{\min}\, v(S_0)/v_{\max} = 11.37$. Theorem 1 gives the boundary; Corollary 3′ (§2.5) gives the ceiling on where the environment can push it.

Our earlier draft tried to compress these three properties into a single equation and a single "law of evolution". That was wrong, and correcting it is the point of this paper. There is no single equation. There are two laws with different fixed points, joined at a switching manifold by an assumption. The structure that matters lives in the *join*, not in either law, and pretending otherwise produced numbers that did not survive their own audit: a headline lifetime ratio that was 3.5x too large, a deadlock boundary whose left tail was made entirely of shock transients, and one self-test that was a tautology.

1.2 What we do instead

We separate the description into four layers and keep them separate.

(i) A formal system. Nine axioms (uptake law, dormant charge law, active accounting law, delivery map, commitment policy, hysteresis switch, environment, death, independence), twelve definitions — including a formal deadlock predicate and a formal phase-transition predicate — and ten inference rules. Every variable carries an explicit domain. A statement that cannot be derived from these is not a theorem in this paper, whatever the simulation shows.

(ii) Four verdicts, not two. A binary success/failure conflates two very different situations: *the work was attempted and nothing landed*, and *the work was attempted and part of the scope does not exist*. The second is not a failure; it is a refusal to answer outside the domain of the question. We write `INVALID` for it, and `NOT_RUN` for the case where no attempt was made. This is not cosmetic. In our sandbox Type-A issues **zero** `FAIL` verdicts in 300 cycles and **708** `INVALID` verdicts, and books every one of them as `PASS` in its own log. That single number is the mechanism of its death.

(iii) Proofs first, simulation second. Theorems are proved from the axioms. The simulation has exactly two roles: to *locate* a boundary that a theorem predicts, and to run an ablation that no theorem covers. When the two disagree, the theorem wins and the code is a bug. This happened four times; the verification report documents each one, including the two that changed published numbers.

(iv) An audit of our own assumptions. We list 28 things we assumed without proving, mark the 11 that are load-bearing, and state which theorem collapses when each fails. We regard this table as a contribution rather than a confession: a reader who wants to disagree with our conclusion needs it in order to know where to push.

1.3 Contributions

C1 — A deadlock boundary that is exact, conditional, and checkable

In a dormant state obeying $\dot{A} = Y\phi_v v(S)(1-A) - \phi_{\text{dorm}} m_0$, the asymptotic charge is $A_\infty = 1 - 1/\chi$ with $\chi = Y\phi_v v/(m_0\phi_{\text{dorm}})$. Hence

$$\text{DL} \iff A_\infty < \theta_{\text{wake}} \iff \chi < \frac{1}{1-\theta_{\text{wake}}} \iff \chi_0 < \frac{1}{f\,(1-\theta_{\text{wake}})}, \qquad f = \frac{v}{V_{\max}},$$

where $\chi_0 = \chi(V_{\max})$ is the margin of a hypothetical $V_{\max}$-capacity agent. The two inequalities are the same statement divided by f; quoting one while computing the other is a factor-f error, and the verification report records exactly that error being made in our own code (its §6.2).

The equivalence itself is algebra. What is *not* algebra is the list of conditions under which it is a statement about the system rather than about a frozen parameter:

- **H1 (entry).** $A(0) \le \theta_{\text{wake}}$ when dormancy begins.
- **H2 (quasi-static substrate).** S is approximately constant on the charging time scale $\tau = 1/(m_0\phi_{\text{dorm}}\chi)$.
- **H3 (wake rule).** Waking requires $A \ge \theta_{\text{wake}}$ for κ consecutive cycles.

All three are necessary. Without H1 the theorem is simply false: an agent that enters dormancy at $A(0) > \theta_{\text{wake}}$ with $A_\infty < \theta_{\text{wake}}$ can satisfy the wake rule before it falls below the threshold. We give that counterexample explicitly (§2.5), because the code we audited violated H1 on its very first line of the dormant branch.

Two consequences replace intuitions that are, in our experience, commonly held.

Two different interventions have been run together under the phrase "adding resources", and they have different answers. Keeping them apart is the correction.

Raising the agent's capacity is expensive, not impossible. With $\chi = c_0 W$,

$$\frac{\partial A_\infty}{\partial W} = \frac{1}{c_0 W^2} = \frac{1}{\chi W} > 0, \qquad \lim_{W\to\infty} \frac{\partial A_\infty}{\partial W} = 0.$$

Inside the deadlock region the finite increment $\Delta W/W = \chi_{\min}/\chi - 1$ needed to cross the threshold is strictly positive; an infinitesimal increase changes nothing, and the marginal return decays quadratically. This is Corollary 3, and it is a statement about a capacity the agent may be *given*: it says the escape is affordable but not free. It does not say the escape is impossible, and the earlier draft quoted it as though it did.

Flooding the environment is bounded, and the bound is the whole story. The environment enters through $W = v(S_0 R)\sigma$, so what matters is the shape of v, not a derivative. Because (Ax1) is unimodal,

$$\chi(R) \le \chi_{\max} := c_0 v_{\max} = \frac{c_0 V_{\max}}{1 + 2\sqrt{K_s/K_i}} = 27.78, \qquad \sup_R A_\infty = 1 - \frac{1}{\chi_{\max}} = 0.964.$$

This is Corollary 3′, and it splits the claim into two regimes that the slogan had merged. Above $\theta = 0.964$ no substrate level escapes at all: the ceiling is below the bar, and more substrate past the peak *lowers* the charging rate because the S^2/K_i term dominates. That is the regime of the $\theta = 0.99$ example, where an agent sits pinned with substrate available. Below 0.964 a flood does help, and it rescues exactly the agents with $\chi(t) > \chi_{\min}\, v(S_0)/v_{\max} = 11.37$, i.e. agents whose margin has fallen less than 36% below the boundary. In the reference scenario $\chi(t) \in [14.63, 17.78]$ throughout, so every deadlocked agent satisfies that inequality — which

means the sandbox's deadlocks are *not* substrate-limited, and the earlier draft's claim that they were is withdrawn here rather than argued away.

What survives is the part that is a theorem: the ceiling $\sup_R A_\infty = 0.964$, the location of its maximum at $S = \sqrt{K_s K_i}$, and the fact that past that point additional substrate is harmful. The slogan "the bottleneck is χ, not the resource" is therefore true in a weaker form than we claimed — the resource can move χ by at most the factor $v_{\max}/v(S_0) = 1.56$ — and only in the dormant phase (M1), with H1–H3.

The boundary is not a constant. It is the image of $\chi(t)$: $\theta^*(t) = 1 - 1/\chi(t)$. Under geometric drift the boundary moves left at a rate set by the drift, which is why a boundary measured at one horizon does not transfer to another. In our sandbox $\theta^*(1) = 0.9437$ and $\theta^*(300) = 0.9317$, and the measured half-saturation point is 0.9325. Appendix A.5 shows why the two numbers must agree: the deadlock probability is a lognormal CDF whose argument vanishes at the terminal anchor, so the centre of the transition *is* the anchor for every dispersion and every population size. Only the width is a free parameter, and only the sample median of this seed (1.0277 rather than 1) moves the centre, by 0.0018 in θ. The first cohort of deaths at $\theta \geq 0.94$ occurs at cycle 27 — 17 of 43 at $\theta = 0.94$, 44 of 50 at $\theta = 0.95$ — and corresponds to the day-1 anchor, because those agents' individual $\chi_i = 17.78\, v_i/m_i$ is already below $\chi_{\min}$ before any drift has accumulated. A single number explains both the onset and the halfway point only when it is read as the family $\theta^*(t)$. A paper that quotes one threshold without its horizon quotes an artefact of its stopping rule.

C2 — A no-loop theorem for closed systems, and the withdrawal of a stronger claim

Take $V = S$ as a Lyapunov function in a batch system with no feed. Then $\dot{S} = -v(S)X < 0$ whenever $S > 0$, so $S \to 0$; consequently $\chi \to 0$, $A_\infty \to -\infty$, and A crosses any threshold in finite time. The downward edge of the hysteresis band is autonomous; the upward edge is unreachable. The band is not a loop but a one-way door.

This is a theorem about *this class* of systems, and the qualifier is not rhetorical. The earlier draft claimed that an external upward trigger is necessary for *any* hysteresis loop. That is false: the van der Pol oscillator [7] has a relaxation loop whose slow variable is internal and which needs no external event to complete a cycle. We withdraw the general claim and keep the conditional one. Two further restrictions are worth stating because they are easy to lose: the result needs $X(t) \geq X_{\min} > 0$ (otherwise $v(S)X$ can stall at $S = 0$ before χ collapses), and it says nothing about fed systems, which is precisely where the interesting dynamics live.

C3 — A power theorem for drift detection, stated with its limits

Let $x_n = a - n\Delta_{\text{drift}} + \varepsilon_n$ with ε_n i.i.d., zero mean, variance σ^2. A statistic T is *affine-invariant* if $T(x + c + dn) = T(x)$ for all c, d. Such a statistic has power exactly α against every non-zero drift rate: subtracting the mean and the trend leaves the noise, and the noise distribution does not depend on the drift. This covers OLS residual variance, window-detrended variance, and any statistic that depends only on the *shape* of the series.

Sliding-window variance is not affine-invariant, but its expectation is constant in n:

$$\mathbb{E}s_n^2 = \sigma^2 \frac{w-1}{w} + \Delta_{\text{drift}}^2 \frac{w^2-1}{12}.$$

It can see drift, but it cannot accumulate evidence: for a fixed threshold there is a $\Delta^*_{\text{drift}}(c, w)$ below which power is α, and above which power is one, with no "the longer I look the more certain I become" regime.

The cumulative potential

$$\Phi_n = \sum_{k=1}^{n} (x_{k-1} - x_k) = n\Delta_{\text{drift}} + \varepsilon_0 - \varepsilon_n$$

has expected value $n\Delta_{\text{drift}}$, $\operatorname{Var} \Phi_n = 2\sigma^2$, and signal-to-noise $n\Delta_{\text{drift}}/(\sqrt{2}\sigma) \to \infty$. That is the good news. The honest qualifications are three:

1. Φ_n is an **endpoint difference**. It is impartial to everything between $t = 0$ and $t = n$; a drift that reverses in the middle cancels in Φ but not in a CUSUM.
2. It is **not sufficient** and **not efficient**. Under Gaussian noise the OLS slope has variance $12\sigma^2/[n(n^2-1)]$, which is smaller by a factor of order n than $\operatorname{Var}(\Phi_n/n)$.
3. Its virtue is interpretive, not statistical: Φ_n *is* the remaining headroom, so a threshold on Φ is a statement about the system rather than about a test statistic.

We therefore present C3 as "a detector with power one that has a physical reading", not as "the right detector".

C4 — An audit: 28 assumptions, 11 load-bearing

Table 1 (§2.2) lists the load-bearing set and the theorem each one supports. The full inventory, with the seventeen non-load-bearing entries and the specific failure mode of each, is §3.1 of the verification report. We single out three because they are the ones a referee will press.

- **A8 (failed work earns nothing).** If failed work still earns, Type-A's charge never reaches zero: on the same capacity trajectory it settles at $A_{200} = 0.85$ and the extinction at cycle 107 disappears. The causal chain of the central empirical claim is therefore *drift → capacity below claim → zero delivery → no income (A8) → linear account to zero (A5)*, of which only the first two steps are axioms. The earlier draft attributed the whole chain to the drift. That was a misattribution.
- **A10 (atomicity).** If Type-A can deliver partially, its income at cycle 100 is $0.927C_0$, its net flow is $1.15 \times 0.927 - 0.93 = +0.137$, and it never dies. The cliff is entirely a consequence of atomic commitment.
- **A20 (the unit of the wake gate).** This one is a warning about our own arithmetic: comparing a pulse peak against a cycle mean inflated the reported lifetime ratio from 2.61 to 9.19. The three self-consistent readings are 2.61 / 2.18 / 9.19, and the choice between them is a modelling decision that must be declared, not hidden.

C5 — A controlled ablation that identifies the mechanism

Theorems do not tell us whether the verdict channel is load-bearing. An ablation does. Two arms, same agents, same capacity trajectory, same seed; the only difference is whether Type-A subscribes to the world's verdict.

arm	extinct	delivered	recorded	gap
self-assessed (`self`)	107	7676.9	8871.7	1194.8 (15.56%)
verdict-wired (`r7`)	survives 300	23602.9	23602.9	0

The `r7` arm has no dormancy — Type-A never had any — so it is not a candidate system. It is a counterfactual that isolates the channel, and that is all we claim for it. But it is enough to move the causal attribution from the drift to the missing verdict, because the drift is identical in both arms.

1.4 What we withdrew

Honesty is cheap to assert and expensive to practise, so the withdrawals are listed as a table rather than as a sentence.

claim in the earlier draft	status now	reason
"one unified equation"	**withdrawn**	M1 and M2 have different fixed points and different stability; the join is assumption A7
"multiplicative availability is necessary in cascaded systems"	**withdrawn**	parallel redundancy gives $1-(1-a)^k$; full correlation gives a; min-selection gives $\min_j a_j$
"the cliff at cycle 100 is a phase transition"	**downgraded to conjecture; false in the independent limit**	with independent agents the survival curve is $1-F_T(t)$ for a C^∞ F_T: no non-analytic point, no diverging susceptibility
"Type-B lives 10x longer"	**corrected to 2.61x**	the wake gate compared a pulse peak with a cycle mean
"the deadlock boundary is measured"	**corrected**	one-cycle verdicts on shocked peaks produced a spurious left tail
"adding resources cannot exit a deadlock"	**narrowed to a ceiling**	true only above $\theta = 0.964$; below it a flood rescues every agent in the reference scenario. The old argument used the derivative on the ascending branch, which cannot support a conclusion that only the S^2/K_i term can give
"R7 is coupled into the loop"	**now true, and shown to matter**	the four verdicts were not implemented in the earlier code; they are now, and the ablation changes the outcome

1.5 What would refute us

A paper that cannot be refuted is not a paper. Each result has a specific failure mode, and we state it in the same units we would use to test it.

- **C1** fails if a dormant system with $\chi > \chi_{\min}$ and $A(0) \leq \theta_{\text{wake}}$ fails to wake for reasons other than substrate limitation. It also fails if measured boundaries do not move with the horizon as $\theta^*(t) = 1 - 1/\chi(t)$ predicts.
- **C2** fails if a closed batch system is exhibited with a sustained hysteresis loop. It says nothing about open systems, and we do not extend it to them.
- **C3** fails if an affine-invariant statistic is shown to have power above α against a pure linear drift. It does *not* claim that all variance methods are useless: a detector that models the trend can work, and should be preferred on efficiency grounds.
- **C5** fails if the `r7` arm dies or the `self` arm survives. Both are checked by `--selftest` (test 10), so the claim is machine-falsifiable in the repository, not merely in principle.

1.5.1 Formal verification artifacts

Everything above that is *algebraic* rather than *empirical* has been machine-checked rather than argued. `formal_verification/Core_Theorems.lean` covers, in Lean 4 using core and `Std` only, the three discrete facts the argument rests on:

- **The threshold equivalence of Lemma 4**, which is the algebraic core of Theorem 1: `chi < 1/(1-theta)` and `1 - 1/chi < theta` are the same statement, given `0 < chi` and `theta < 1`. The same section records that the reversal `theta < 1 - 1/chi` is the *escape* condition rather than the deadlock condition, and exhibits a machine-checked counterexample at the paper's own operating point (`chi = 17.7778`, `theta = 0.95`) for the reversed form.
- **The mutual exclusion of the two modes**, in both readings: the wake and dormancy threshold predicates cannot both fire at one charge, and the transition function returns one constructor. The second reading is a property of the machine rather than of the charge, which is why the band `theta_dorm < A < theta_wake` carries memory instead of behaving as a coin flip.
- **The four-valued verdict as a discrete collapse**: the map from a physical witness to `PASS` / `FAIL` / `INVALID` / `NOT_RUN` is total, deterministic and surjective, and on the delivered axis it is a step with no intermediate value.

The file compiles with zero errors, zero warnings and no `sorry`, and its axiom footprint is printed from the source by `#print axioms` rather than summarised here: two of the results name no axioms at all, one names only `propext`, and the rest name `propext`, `Classical.choice` and `Quot.sound`, which is the standard footprint of rewriting with `Std`'s ordered-field lemmas. Two further `native_decide` axioms appear only for the closed arithmetic of the counterexample.

We provide this artifact as a verifiable baseline rather than as a claim that the argument is beyond review, and we note one boundary so that it cannot be mistaken for a result: the file does not formalise the differential equations, the order-statistic law or the simulation. It checks the discrete skeleton and nothing above it. Independent scrutiny and correction are welcome, and we would rather be told where the formalisation and the text disagree than have it inferred that they cannot.

1.6 Relation to existing work

The ingredients are classical and we claim no novelty for them individually: Michaelis–Menten kinetics and substrate inhibition for the uptake law [5]; the energy-charge account of adenylate pools for the charge variable [1]; Lyapunov and LaSalle for the no-loop result; AIMD for the adjustable-scope arm; the endpoint-difference/CUSUM family for Φ; order statistics for the width of the transition; and the reliability-theory product law for the cascaded case. The contribution is the assembly: keeping two laws separate, giving the verdict four values with explicit scope semantics, proving the boundary as a conditional rather than quoting it as a law, and publishing the assumption audit that most frameworks keep in a drawer.

The agent-side literature is recent and bears directly on the verdict, so we state the relation rather than leave it implied. Model-based judges carry position, verbosity and self-enhancement biases [8], which is why a checker drawn from the same distribution as the agent is not an independent witness. Intrinsic self-correction — asking the model to grade its own output without new external evidence — does not reliably improve reasoning [3, 6]. Agent benchmarks that score against a real artefact, rather than against a self-report, are the practical form of the witness requirement [4]. Multi-agent failures cluster in verification and in inter-agent misalignment [2], which is the taxonomy-level version of the case this paper isolates. The four-valued verdict is the audit-side counterpart: it refuses to fold "we did not look" and "it did not run" into "it failed", because those three states assign responsibility to three different parties.

Two named ideas are used as objects of study rather than as citations. Goodhart's observation — that a measure ceases to be a good measure when it becomes a target — is the informal version of the A8/A10 mechanism: the agent optimises the recorded quantity, and the recording has been decoupled from delivery. The microbiology of persistence supplies the

phenomenology of the dormant law (cheap upkeep, saturating recharge); we use the shape and make no empirical claim about any organism.

1.7 Organisation

Section 2 fixes the formal system: domains, twelve definitions, the ten inference rules, the two-regime model with the proof that M1 and M2 are inequivalent, the proof of Theorem 1 including the three conditions under which it is a statement about the system rather than about a frozen parameter, the atomic mechanism that places the onset at cycle 100, and the ledger semantics of the four verdicts. Section 3 reports the sandbox: the base scenario, the lifespan gap, the ledger discrepancy, the boundary sweep with its two analytic anchors, the ablation, and the sensitivity tables. Section 4 discusses the three consequences of the framework and states what it does not cover; Section 5 concludes.

Appendix A proves the four lemmas on which the quantitative claims depend — the non-existence of a singular limit between M1 and M2 (A.7, placed first), the closed form for the wake-gate crossing (A.1), the stranding latency (A.2) and the order-statistic law of the boundary (A.5) — and records the saturation assumption A28 with its dependency chain (A.28).

The verification report that preceded this draft — including the four defects it found and the numbers that changed because of them — is kept in the repository as a separate document. We cite it rather than paraphrase it, because the corrections are part of the result.

The eleven load-bearing assumptions are collected in Table 1 (§2.2), with the axiom or decision that realises each and the result that fails if it is false. The remaining seventeen are inventoried in §3.1 of the verification report.

2. Method

2.0 Naming conventions and the status of every symbol

Two naming conventions are fixed here and used without exception in Sections 2–5.

M1 / M2. The dormant and active phases are governed by *different* equations, and the paper refers to them as separate models:

label	phase	equation	character
M1	dormant	$\dot{A} = Y\phi_v v(S)(1-A) - \phi_{\text{dorm}} m_0$	saturating, negative feedback $(1-A)$
M2	active	$\dot{A} = \frac{m_0}{D_0}\left[(1+\epsilon_m)d - c\right]$	linear accounting, no feedback

The earlier draft used "M1" for the active branch and "M2" for the dormant one. That labelling is reversed here, and it is reversed because the substantive claim is that the two are *not* the same model: the labels follow the order in which the results depend on them, not the order in which they appear in a cycle. Any citation of the earlier draft must be read with this swap.

Ax / A. Axioms are written `Ax1`–`Ax9`. Implicit assumptions inventoried by the audit are written `A1`–`A28`. The two numbering schemes are unrelated: `A5` (an assumption about the active law) is realised by `Ax3` (the axiom), not by `Ax5` (the commitment policy). Table 1 is the only place where both numbering schemes appear in the same row; the verification report (§3.1) gives the full inventory, including the seventeen entries that are not load-bearing.

Quantities. C is the cycle-mean capacity (what an agent can deliver over a cycle); W is the pulse peak (the rate at which a dormant agent charges). They differ by a factor of 1.74 in the reference scenario and are never interchanged. χ is the charging margin; χ_0 is χ evaluated at $W = V_{\max}$, and $f = W/V_{\max}$. The two are routinely confused in the numbers, so we fix the reference values once: $c_0 = Y\phi_v/(m_0\phi_{\text{dorm}}) = 5.5556$, $\chi_0 = c_0 V_{\max} = 55.556$, and at the operating point $W(1) = v(S_0) = 3.2$ one has $f = 0.32$ and $\chi(1) = f\chi_0 = 17.7778$. A quoted margin is meaningless without saying which of the two it is.

Two symbols that are not the same symbol. Delta appears twice with unrelated meanings, written differently so that a reader who meets one cannot silently substitute the other. δ is the *upkeep deficit*: maintenance owed but not paid between the saturation reference and the onset, in units of $m_{0,i}$ (Corollary 4, §2.6). Δ_{drift} is the *linear drift rate*: the per-step decline of the mean under the C3 null (Proposition 5, §2.8). The first is a stock, bounded by the horizon; the second is a rate. The earlier draft wrote both as δ, separating them only by context; the subscript is the fix.

Three epsilons, two glyphs. The letter carries three unrelated roles, and the glyph together with the subscript is what separates them. ϵ_m (lunate ϵ, subscript m) is the *accounting markup* of Ax3: a delivered unit is credited at $1+\epsilon_m$ times its own capacity against a maintenance charge of one, and $\epsilon_m = 0.15$ in §2.9. It is what sets the duty-cycle threshold $\epsilon_m/(1+\epsilon_m) = 13.04\%$ of §3.7. ε (bare) is the *drift rate* of Ax7: the fractional decrement of R per ten-cycle block, $\varepsilon = 0.01$, applied multiplicatively so that $R = (1-\varepsilon)^k$ after k blocks. Its logarithmic equivalent is $-\ln(0.99) = 0.01005$, not 0.01; the paper uses the multiplicative form throughout, and every number in §3.5 is computed from it. ε_n and ε_0 (subscripted without m) are the *observation noise* of the C3 null in Proposition 5, i.i.d. with zero mean and variance σ^2. Markup and drift differ by a factor of fifteen and never appear in the same expression — Ax3 and its corollaries against Ax7 and A.1 — so substituting one for the other produces a visible inconsistency rather than a rounding error. That is the whole reason they are printed with different glyphs; it is not a claim that they are related.

κ, κ_i, and A_{sleep}. Three run lengths govern the switch, and none of them is a charge. κ_i is a *state*: the agent's current run of consecutive cycles that satisfy the gate it is under (§2.1). κ without a subscript is the *constant* a rule requires, $\kappa = 3$ in §2.9, and the same length governs both waking (Ax6, H3) and the deadlock confirmation of I5, because one shocked or borderline sample is evidence in neither direction; the counters are separate and start at zero. A_{sleep} is not a charge despite the letter: it is the number of consecutive `FAIL` verdicts that triggers dormancy (Ax6, A.2), equal to 2 in §2.9. The earlier draft wrote that clause $f \geq 2$ with $f = W/V_{\max}$; since $W \leq V_{\max}$ the predicate could never fire, and it was never what the implementation tested.

2.1 State space and domains

Time is discrete, $t \in \mathbb{N}$, one step per fed-batch cycle. The state of agent i is

$$x_i(t) = \big(A_i,\, m_i,\, c_i,\, d_i,\, \kappa_i\big),$$

with domains

$$A_i \in \mathbb{R}, \qquad m_i \in \{\text{act}, \text{dorm}, \text{dead}\}, \qquad c_i \in [0, C_{0,i}], \qquad d_i \in [0, c_i], \qquad \kappa_i \in \mathbb{N}_0.$$

Two domain conventions are adopted deliberately and they constrain what can be claimed later.

1. A **is not restricted to** $[0,1]$**.** `Ax3` is linear and does not preserve the unit interval. The implementation projects the upper bound ($A \leftarrow \min(A, 1)$) and treats $A \leq 0$ as an absorbing event. A projection is not an invariant, so the model is an affine system with an

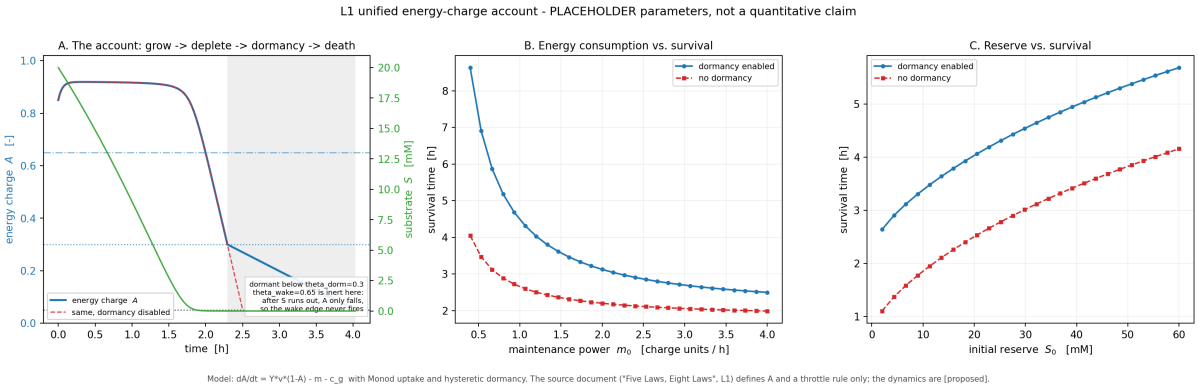


Figure 1: The L1 energy-charge account. (a) Growth, substrate depletion, dormancy and death, with dormancy enabled (solid) and disabled (dashed). (b) Survival against maintenance rate. (c) Survival against initial reserve. Open markers are censored, not death times.

absorbing set, not a flow on a compact simplex. Lemma 1 is stated for the branch $A < 1$, which is the only branch Theorem 1 uses.

2. $A \leq 0$ **is death, not equilibrium.** `Ax8` makes it irreversible. There is therefore no interior equilibrium of `M2` with $d = 0$; the only fixed point of the active law with zero delivery lies on the boundary.

The environment state is (R, σ), with $R > 0$ the resource index and $\sigma \in \{\sigma_s, 1\}$ the shock multiplier. Capacity and peak are

$$C(t) = \frac{1}{n}\sum_{j=0}^{n-1} v\big(S_0 R(t)(1-\delta_s)^j\big)\sigma(t), \qquad W(t) = v\big(S_0 R(t)\big)\sigma(t),$$

with $n = 8$, $\delta_s = 0.22$, $S_0 = 0.5$. In the reference scenario $C(1) = 1.8426$ and $W(1) = 3.2000$.

Figure 1 fixes the vocabulary of the rest of the paper: the charge account of a single agent with dormancy enabled and disabled, and the two surfaces against which the reserve is read.

2.2 Axioms

The following nine statements are the entire starting set. Anything not derivable from them is labelled empirical.

Ax1 (uptake). $v(S) = V_{\max} S/(K_s + S + S^2/K_i)$, with $V_{\max} = 10$, $K_s = 1$, $K_i = 4$. $v \in C^\infty$, $v(0) = 0$, v is increasing on $[0, \sqrt{K_s K_i}) = (0, 2)$ and decreasing beyond. The operating point is $S_0 = 0.5 < 2$, so C increases in R throughout the scenario. Figure 2 shows the law and its inhibitory branch.

Ax2 (dormant charging, M1). For $m = \text{dorm}$, $\dot{A} = Y\phi_v v(S)(1-A) - \phi_{\text{dorm}} m_0$. This is the law on the shock-free branch. During an episode the uptake is $W = \sigma_s v(S)$, and the same equation holds with $v(S)$ replaced by W; because D8 defines χ on the peak W, the identity $A_\infty = 1 - 1/\chi$ is the fixed point of this equation exactly when $W = v(S)$, which is the branch on which Theorem 1's deadlock verdicts are issued (I5).

Ax3 (active accounting, M2). For $m = \text{act}$, $\dot{A} = \dfrac{m_0}{D_0}[(1+\epsilon_m)d - c]$.

Ax4 (delivery map). $d = c \cdot \mathbf{1}[C \geq c]$ for Type-A (atomic); $d = \min(c, C)$ for Type-B (divisible).

Ax5 (commitment policy). $c_A \equiv D_0$; $c_B = a$ with $a_{t+1} = \min(D_0, 1.05 a_t)$ on `PASS` and $a_{t+1} = 0.95 a_t$ on `FAIL`.

Ax6 (hysteresis switch; Type-B). For an agent that may enter dormancy (Type-B; I4 excludes Type-A), act $\to$ dorm iff $A < \theta_{\text{dorm}}$ or A_{sleep} consecutive cycles were judged `FAIL`;

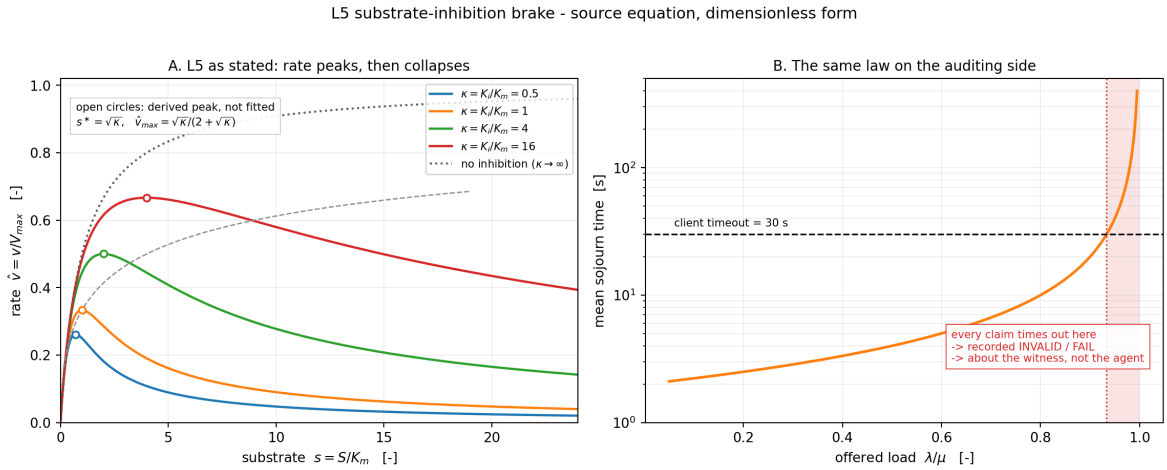


Figure 2: The uptake law and its inhibitory branch. (a) The rate rises to a derived maximum at the geometric mean of the half-saturation and inhibition constants, then collapses as the reciprocal of the substrate; open circles mark the analytic peak, the dashed curve the locus of achievable peaks. (b) The same overload logic on the auditing side: mean sojourn time in an M/M/1 queue.

dorm $\to$ act iff $A \geq \theta_{\text{wake}}$ for κ consecutive cycles. The second entry condition is the verdict streak of §2.7, not a load ratio; the implementation keeps it as `SLEEP_AFTER=2`.

Ax7 (environment). R is multiplied by $1-\varepsilon$ every 10 cycles, so that $R(t) = (1-\varepsilon)^{\lfloor t/P \rfloor}$; $\sigma = \sigma_s$ on cycles t mod $P \in \{0, \ldots, L-1\}$ and 1 otherwise.

Ax8 (death). $A \leq 0 \Rightarrow m =$ dead, irreversibly.

Ax9 (independence). Agents share the exogenous $C(t)$ and do not interact. v_i and $m_{0,i}$ are independent static lognormal factors with $\sigma_v = 0.08$, $\sigma_m = 0.12$.

Inference rules. Ten rules are used, and a statement derived with anything else is not a proof in this paper: modus ponens (IR1); universal instantiation, agent to population (IR2); induction, used for the cascaded product (IR3); monotone convergence, used for $A(t) \to A_\infty$ (IR4); comparison principle and Gronwall, used for the quasi-static error (IR5); l'Hôpital and asymptotic expansion, used for the tail of v (IR6); reductio, used for the no-loop theorem and for the empty-set argument of §2.7 (IR7); Lyapunov functions, used for the closed batch system (IR8); the law of large numbers, used for the $N \to \infty$ reading of the boundary (IR9); and the delta method, used for the interval on τ (IR10). A numerical agreement is not a proof, a parameter sweep is not a proof, and a parameter chosen to produce a phenomenon is not a derivation: any statement that rests on those three is recorded as an empirical regularity.

Table 1. The eleven load-bearing assumptions of the audit, and what each supports.

ID	assumption in words	axiom / decision that realises it	result that fails if it is false
A1	uptake is algebraic, not stateful	Ax1	χ undefined; Theorem 1 not computable
A5	the active phase is linear with no feedback	Ax3 (M2)	recovery to $A \approx 0.95$; no extinction
A7	M1 and M2 may be joined at the switch	Ax6	two theorems about two systems; the join is unjustified
A8	failed work earns nothing	Ax3 with $d = 0$	extinction disappears
A10	Type-A tasks are atomic	Ax4	positive margin at cycle 100; no step

ID	assumption in words	axiom / decision that realises it	result that fails if it is false
A13	no reproduction, inheritance or variation	Ax9	cohort survival, not evolution
A14	agents are independent	Ax9	frequency dependence; a genuine phase transition possible
A16	utilisation is fixed	scenario choice	ordering reverses near $U = 0.445$
A20	the wake gate is peak-to-peak	wake rule, §2.7	$279 \to 233$ or 983
A21	deadlock needs three shock-free samples	§2.7, I5	spurious left tail at $\theta = 0.90$
A27	all eleven kinetic constants are placeholders	parameter table	numbers are dimensionless shapes, not predictions

2.3 Definitions

Twelve definitions are used. They are numbered to agree with the formal verification report, so that a reader can move between the two documents without re-indexing.

D1 (delivery and commitment). $d_i(t)$ is the work that actually landed in cycle t; $c_i(t)$ is the scope the agent committed to before the cycle began.

D2 (availability). $\mathrm{Avail}_i(t) = d_i(t)/c_i(t)$ for $c_i(t) > 0$, with the convention $0/0 = 0$. This makes $\mathrm{Obs}_i = \mathrm{Cap}_i \cdot \mathrm{Avail}_i$ an identity rather than a theorem, which is why availability cannot be used as independent evidence about capacity.

D3 (observation). $\mathrm{Obs}_i(t) = \mathrm{Cap}_i(t) \cdot \mathrm{Avail}_i(t)$. Type-A records $\mathrm{Cap}_i = c_i$, so its observation is the claim; Type-B records $\mathrm{Cap}_i = d_i$, so its observation is the delivery.

D4 (deadlock). $\mathrm{DL}_i :\iff m_i = \mathrm{dorm} \ \wedge \ \lim_{t\to\infty} A_i(t) < \theta_{\mathrm{wake}} \ \wedge \ \exists S > 0 : v(S) > 0$. The third clause separates deadlock (resources present, charge unreachable) from starvation (resources absent).

D5 (stranding). $\mathrm{ST}_i :\iff m_i = \mathrm{dorm} \ \wedge \ \forall t \geq t_0 : W_i(t) < w_h$, $w_h = 0.85\, W_i(1)$. DL is a statement about the limit of A; ST is a statement about the supply side. Neither implies the other.

D6 (extinction). $E_A(t) :\iff \#\{i : \ \mathrm{kind}_i = A, \ m_i \neq \mathrm{dead}\} = 0$.

D7 (collapse). Operational definition used by the code: with b the median of d_A over the first 20 cycles, $t_{\mathrm{collapse}} = \max\{t : d_A(t) \geq b/2\} + 1$. This is a definition, not a theorem; a different threshold gives a different cycle.

D8 (charging margin). $\chi_i(t) = Y\phi_v W_i(t)/(m_0\phi_{\mathrm{dorm}})$ and $\theta_i^*(t) = 1 - 1/\chi_i(t)$. With the $V_{\max}$ normalisation $\chi_{0,i} = Y\phi_v V_{\max}/(m_0\phi_{\mathrm{dorm}})$ and $f_i = W_i/V_{\max}$, one has $\chi_i = f_i\chi_{0,i}$.

D9 (cumulative causal potential). $\Phi_n = S_0 - \hat{S}_n = \sum_{k\leq n}(\hat{S}_{k-1} - \hat{S}_k)$.

D10 (hysteresis loop). A state sequence that returns repeatedly to $[\theta_{\mathrm{dorm}}, \theta_{\mathrm{wake}}]$ with the upward and downward edges triggered by conditions of different type (one autonomous, one exogenous). We flag here that this definition writes *asymmetry* into the definiens, so that "asymmetry is necessary for a loop" is a tautology under D10 and false under the standard dynamical-systems reading; §2.8 uses the standard reading, which is why the general claim is withdrawn.

D11 (phase transition). An order parameter $s(g)$ is non-analytic at some g_c, or the susceptibility $\chi = -\partial s/\partial g$ diverges there, and the behaviour survives $N \to \infty$. This is the strict definition used to reject the phase-transition reading in §3.5.

D12 (functional). Alive and not stranded: $m \neq \mathrm{dead} \ \wedge \ \neg\mathrm{ST}$. This is the ordinate of panel 1 of the sandbox figure in §3.1.

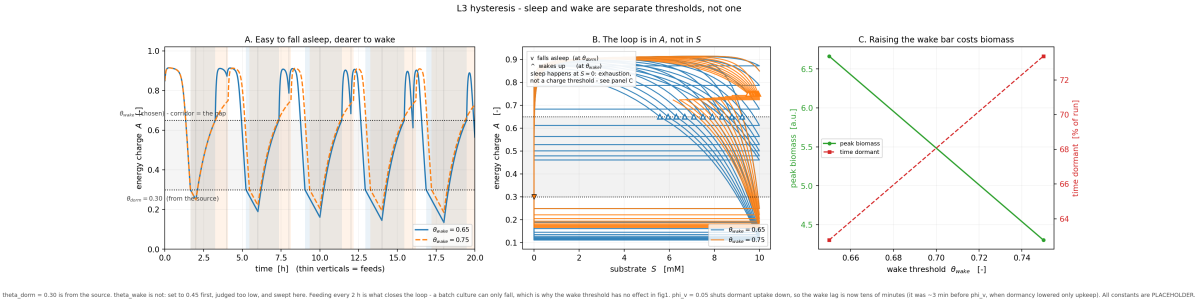


Figure 3: Hysteresis in a 2 h feeding regime. (a) Charge over time. (b) The (S, A) plane: the loop lives in A while S returns to zero every cycle. (c) Peak biomass against the wake threshold.

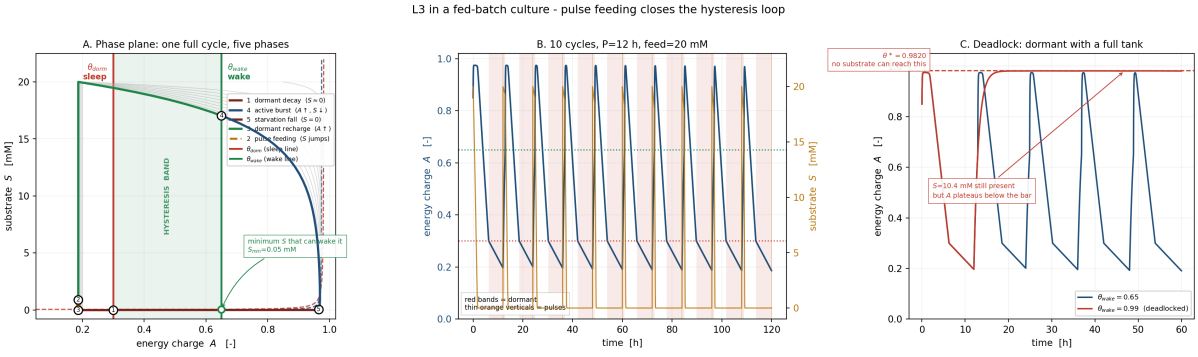


Figure 4: Fed batch: a closed five-phase limit cycle. (a) Phase plane with one cycle highlighted. (b) Time series with dormant bands and pulses. (c) Deadlock at a wake threshold of 0.99: the agent is pinned below the bar with substrate still available.

2.4 The two-regime model

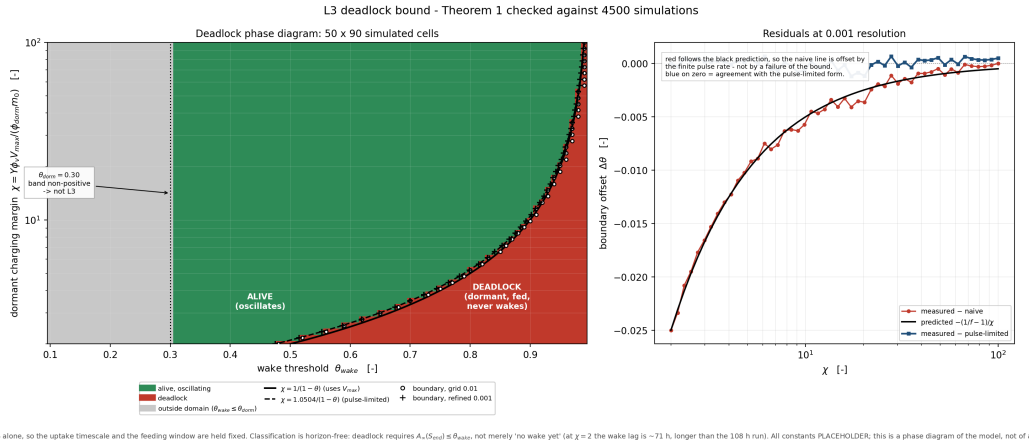
Figure 3 shows the phenomenon the switch `Ax6` is there to represent: in a periodic feeding regime the charge returns to the band every cycle while the substrate does not, so the loop lives in A and not in S. Figure 4 shows the same structure in a closed fed batch, where the cycle is a genuine limit cycle and the deadlock at $\theta_{\text{wake}} = 0.99$ is visible as an agent pinned below the threshold with substrate still in the vessel.

Proposition 1 (M1 and M2 are not equivalent). Fix $v > 0$ and write $a = Y\phi_v v > 0$, $b = \phi_{\text{dorm}} m_0 > 0$. `M1` has the interior fixed point $A^*_{\text{M1}} = 1 - b/a = 1 - \phi_{\text{dorm}} m_0/(Y\phi_v v)$, which is exponentially stable with rate $a = Y\phi_v v$. `M2` with $d = 0$ has no interior fixed point and decreases at the constant rate $m_0 c/D_0$; with $d = c$ it increases without bound until the projection acts. The two have opposite qualitative behaviour for every $v > 0$.

The fixed point is the same expression as $A_\infty = 1 - 1/\chi$ of Lemma 1, because D8 defines $\chi = Y\phi_v v/(\phi_{\text{dorm}} m_0) = a/b$; a reader who meets $1 - 1/\chi$ in §2.5 and $1 - b/a$ here is reading one formula, not two.

Proof. $\partial\dot{A}/\partial A = -Y\phi_v v = -a < 0$ for `M1`, so the deviation from the fixed point decays as e^{-at}; $\partial\dot{A}/\partial A = 0$ for `M2`, which is therefore affine in A and has no interior fixed point when $d = 0$. □

Corollary 1 (the 107-cycle extinction belongs to M2). On the reference capacity trajectory `M1` predicts $A^* = 1 - \phi_{\text{dorm}} m_0/(Y\phi_v v) = 1 - 0.018/0.32 = 0.94375$ at the day-1 peak $W(1) = v(S_0) = 3.2$, and 0.9402 at $W(100) = 3.008$, both above $A(0) = 0.85$; hence A would rise toward the fixed point with $\tau = 3.125$ cycles and would not fall. The largest shock in the record subtracts at most $Lm_0 = 0.30$, about a third of the fixed point, and even at the largest sampled $m_{0,i} = 0.183$ the subtraction is $2m_{0,i} = 0.37$: A stays above $\theta_{\text{dorm}} = 0.30$, so no shock drives it into dormancy, and the loss is restored within $3\tau \approx 9$ cycles. The onset at cycle 100



χ scanned by varying ϕ_r alone, so the uptake timescale and the feeding window are held fixed. Classification is horizon-free: deadlock requires $A_\infty(S_{end}) \le \theta_{wake}$, not merely 'no wake yet' (at $\chi = 2$ the wake lag is ~71 h, longer than the 108 h run). All constants PLACEHOLDER; this is a phase diagram of the model, not of any measured system.

Figure 5: The deadlock phase diagram over the wake threshold and the charging margin (4500 cells). The black line is Theorem 1 in its maximum-rate form, the dashed line the pulse-corrected form. Right panel: boundary offsets against the margin, showing the measured deviation tracking the predicted offset rather than sitting on zero.

lies seven cycles after the episode 92–93 and fifteen before the next, so no shock is inside the decisive window. `M2` with $d = 0$ decreases at m_0 per cycle and crosses zero in $A(100)/m_0 \approx 4.6$ cycles. The extinction is a property of the accounting law, not of the metabolic law.

This proposition is the reason the paper does not present a unified equation. `Ax6` joins the two regimes; the join is assumption A7, and it is an assumption because no limit of one law produces the other at the switching manifold: M1 has a stable interior point where M2 has none, and M2 has a conserved linear drift where M1 has exponential relaxation.

2.5 Theorem 1: the deadlock boundary

Figure 1 gives the charge account that Lemma 1 formalises: growth, substrate depletion, dormancy and death for the same strain with dormancy enabled and disabled, together with survival against maintenance rate and against initial reserve. Figure 5 is the phase diagram of the resulting boundary in the $(\theta_{\text{wake}}, \chi)$ plane, with the analytic line of Theorem 1 drawn on it.

Lemma 1 (closed form). If S is fixed on the charging time scale, `M1` has

$$A(t) = A_\infty + \big(A(0) - A_\infty\big)e^{-t/\tau}, \qquad A_\infty = 1 - \frac{1}{\chi}, \qquad \tau = \frac{1}{m_0 \phi_{\text{dorm}} \chi}.$$

Proof. Write `Ax2` as $\dot{A} = -(A - A_\infty)/\tau$ and integrate. □

Lemma 2 (monotonicity). $A(t)$ is monotone between $A(0)$ and A_∞ and does not oscillate. *Proof.* The error term is an exponential with a fixed sign. □

Lemma 3 (no crossing). If $A(0) \le \theta$ and $A_\infty < \theta$, then $A(t) < \theta$ for all $t > 0$. *Proof.* By Lemma 2, $A(t)$ lies between $A(0)$ and A_∞; if $A(0) = \theta$ the approach is strict. □

Lemma 4 (threshold equivalence). For $\theta \in (0, 1)$ and $\chi = f\chi_0$,

$$A_\infty < \theta \iff \chi < \frac{1}{1-\theta} \iff \chi_0 < \frac{1}{f(1-\theta)}.$$

Proof. $1 - 1/\chi < \theta \iff \chi < 1/(1-\theta)$; substitute $\chi = f\chi_0$. □

Theorem 1. Assume

- **H1 (entry):** at the first dormant cycle, $A(0) \le \theta_{\text{wake}}$;
- **H2 (quasi-static):** S is constant to within the tolerance of Lemma 1 on the scale τ;
- **H3 (wake rule):** waking requires $A \ge \theta_{\text{wake}}$ for κ consecutive cycles.

Then

$$\text{DL} \iff A_\infty < \theta_{\text{wake}} \iff \chi < \frac{1}{1-\theta_{\text{wake}}} =: \chi_{\min} \iff \chi_0 < \frac{1}{f\,(1-\theta_{\text{wake}})} = \frac{\chi_{\min}}{f}.$$

Proof. ($\Leftarrow$) By H1 and Lemma 3, $A(t) < \theta_{\text{wake}}$ for all $t \geq 0$, so H3 never fires; by D4, deadlock. ($\Rightarrow$) Suppose $A_\infty > \theta_{\text{wake}}$. By Lemma 2, A approaches A_∞ from below. If $A(0) \leq \theta_{\text{wake}}$, there is a finite

$$t_c = \tau \ln \frac{A_\infty - A(0)}{A_\infty - \theta_{\text{wake}}}$$

with $A(t_c) = \theta_{\text{wake}}$; after κ further cycles H3 is satisfied. If $A(0) > \theta_{\text{wake}}$ the threshold is met immediately. Hence no deadlock. Lemma 4 gives the equivalence chain. □

The critical case is a definition gap, not a theorem. At $\chi = \chi_{\min}$ we have $A_\infty = \theta_{\text{wake}}$; A approaches the threshold strictly from below and never attains it, so H3 never fires while D4 (strict inequality) does not classify the agent as deadlocked. The interval $\chi = \chi_{\min}$ is therefore a measure-zero set on which the agent is functionally dead and formally not deadlocked. Both the theorem and the code use the strict reading.

Corollary 2 (the boundary has a horizon). Because χ is a function of the current peak,

$$\theta^*(t) = 1 - \frac{1}{\chi(t)} = 1 - \frac{1}{c_0 W(t)}, \qquad c_0 = \frac{Y\phi_v}{m_0\phi_{\text{dorm}}}.$$

Under geometric drift W decreases, so θ^* decreases. A threshold quoted without its horizon is the boundary of a stopping rule, not of the system. Numerically, $\theta^*(1) = 0.9437$ and $\theta^*(300) = 0.9317$.

Corollary 3 (marginal capacity). With $\chi = c_0 W$,

$$\frac{\partial A_\infty}{\partial W} = \frac{1}{c_0 W^2} = \frac{1}{\chi W} > 0, \qquad \frac{\partial A_\infty}{\partial W} \to 0 \;\; (W \to \infty).$$

Inside the deadlock region, restoring $A_\infty > \theta_{\text{wake}}$ requires the finite relative increment $\Delta W/W = \chi_{\min}/\chi - 1 > 0$. An infinitesimal increase changes no verdict. The statement is local and it is about W as a free variable; W is not free, and the next corollary is what that costs.

Corollary 3′ (the uptake peak caps the boundary). With $W(R) = v(S_0 R)\sigma$, $\sigma \leq 1$, and (Ax1),

$$\chi(R) = \chi_0 \frac{v(S_0 R)}{V_{\max}}\,\sigma \;\leq\; \chi_0 \frac{v_{\max}}{V_{\max}} = \frac{\chi_0}{1 + 2\sqrt{K_s/K_i}} \;=:\; \chi_{\max},$$

with equality iff $S_0 R = \sqrt{K_s K_i}$ and $\sigma = 1$. Consequently

$$\sup_{R>0} A_\infty(R) = 1 - \frac{1}{\chi_{\max}} = 0.964, \qquad \sup_{R>0} \theta^*(R) = 1 - \frac{1 + 2\sqrt{K_s/K_i}}{\chi_0} = 0.964.$$

Proof. Differentiating (Ax1), $v'(S) = V_{\max}(K_s - S^2/K_i)/(K_s + S + S^2/K_i)^2$, which vanishes exactly at $S = \sqrt{K_s K_i}$ and changes sign there; at that point $v_{\max} = V_{\max}/(1 + 2\sqrt{K_s/K_i})$. Substituting $\chi = c_0 W$ with $\chi_0 = c_0 V_{\max}$ gives $\chi_{\max}$, and Lemma 4 gives the boundary. □

Three consequences, and the third is a retraction. First, above $\theta = 0.964$ no substrate level can restore an agent: the ceiling is below the bar, and past the peak more substrate *lowers* χ. This is the regime of the $\theta = 0.99$ example. Second, below 0.964 a flood does help, and it rescues exactly the agents with $\chi(t) > \chi_{\min} v(S_0)/v_{\max} = 11.37$; in the reference scenario $\chi(t) \in [14.63, 17.78]$, so every deadlocked agent qualifies. Third: the claim that adding resources cannot exit a deadlock is therefore false in the range this paper's sweep covers, and we withdraw

it in that range. What the inhibition term rules out is not escape but *unbounded* escape — and that is the difference between the slogan we published and the theorem we can prove.

Remark on H1. H1 is not a technicality. If dormancy is entered through the verdict clause (two consecutive `FAIL` cycles) rather than through low charge, then $A(0)$ can exceed θ_{wake} while $A_\infty < \theta_{\text{wake}}$. Take $A(0) = 0.9$, $\theta_{\text{wake}} = 0.65$, $A_\infty = 0.5$: the crossing time is $t_c = \tau \ln(0.4/0.15) = 0.98\tau$, so for $\kappa \le 0.98\tau$ the agent wakes before it falls below the threshold and is not deadlocked. A verdict issued on the first dormant cycle is therefore unsound. The implementation this paper supersedes did exactly that; §2.7, invariant I5, states the replacement.

2.6 The atomic mechanism

Theorem 1 concerns the dormant phase. The extinction in the base scenario occurs in the active phase, and its mechanism is separate and purely algebraic.

Proposition 2 (delivery onset). Let $\rho(t) = C(t)/C_0$ and let the Type-A claim be $c_i = UC_{0,i}$, so that $c_i/C_i(t) = U/\rho(t)$ for every agent i. Then $d_i(t) = c_i$ iff $\rho(t) \ge U$, and $d_i(t) = 0$ otherwise. Hence the first cycle with $d_i = 0$ is $\min\{t : \rho(t) < U\}$, independent of i and of v_i.

Proof. Immediate from `Ax4` and the proportionality of claim and capacity to v_i. □

Proposition 3 (drought rate). The implemented form of `Ax3` is $\Delta A_i = m_{0,i}\big[(1+\epsilon_m)d_i - c_i\big]/D_{0,i}$, i.e. the prefactor is the per-agent upkeep rate $m_{0,i}$ itself. While $d_i = 0$ and $c_i = D_{0,i}$ this gives $\Delta A_i = -m_{0,i}$ exactly. Hence the time from onset to death is $A_i(t_c)/m_{0,i}$ and the death cycle is $t_c + \lceil A_i(t_c)/m_{0,i} \rceil$.

Proof. Substitute $d = 0$, $c = D_0$ in `Ax3`. □

Propositions 2 and 3 give the two observables of the base run. With $U = 0.93$ and $\rho(99) = 0.9342$, $\rho(100) = 0.9271$, the onset is exactly cycle 100, and no agent dies on that cycle because $A_i(100) > 0$. With $A_i(100) \in [0.6158, 0.7558]$ and $m_{0,i} \in [0.1162, 0.1829]$, the deaths fall in cycles 104–107, four to seven cycles after the onset, entirely inside a shock-free interval (the nearest shocks are at 92–93 and 115–116). The spread is not noise: it is the heterogeneity of $m_{0,i}$, which is the only parameter that differs across Type-A agents in this equation.

Corollary 4 (drought clock). Suppose that at some cycle t_s every Type-A agent is at the projection $A = 1$, and let

$$\delta = \sum_{s=t_s+1}^{t_c} \Big(1 - \epsilon_m \, \mathbf{1}[\text{PASS at } s]\Big)$$

be the deficit accumulated between t_s and the onset, measured in units of $m_{0,i}$. Then $A_i(t_c) = 1 - \delta m_{0,i}$ and

$$t_{\text{death},i} = t_c + \left\lceil \frac{1}{m_{0,i}} - \delta \right\rceil .$$

Proof. Each cycle adds $+\epsilon_m m_{0,i}$ on `PASS` and $-m_{0,i}$ otherwise, and the projection is inactive after t_s by hypothesis. Sum and divide by $m_{0,i}$. □

In the base run the last cycle at which every Type-A agent is at the projection is 91; the deficit to cycle 100 is $\delta = 3 - 0.15 \times 6 = 2.1$, so $A_i(100) = 1 - 2.1\, m_{0,i}$ exactly. The deaths predicted by the corollary are distributed over cycles 104–107 with multiplicities $11, 23, 13, 3$, which is the observed distribution. The four-cycle spread of the extinction is therefore a deterministic function of the $m_{0,i}$ sample, not a stochastic effect.

The quantity the corollary makes available to an operator is the interval between the saturation reference and the death, $t_{\text{death}} - t_c = \lceil 1/m_{0,i} - \delta \rceil \in [4, 7]$ cycles. It is worth separating this from the raw headroom $1/m_{0,i} \in [5.47, 8.61]$: the offset between them is the deficit $\delta = 2.1$, already spent before the clock starts, and only the reduced interval is a lead time that can be promised to anyone.

Assumption A28 (pre-shock saturation). Every Type-A agent attains $A = 1$ under the projection before the first shock cycle.

Corollary 4 takes the saturation reference at $t_s = 91$ and treats δ as a constant; that step presumes the agents saturated before the first shock, which is not implied by the axioms. The presumption is stated here rather than in passing, and it is verified rather than assumed: on the `PASS` branch, $\Delta A_i = \epsilon_m m_{0,i}$, so $t_{\text{sat},i} = \lceil (1 - A_0)/(\epsilon_m m_{0,i}) \rceil = \lceil 1/m_{0,i} \rceil \in [6, 9]$ for $m_{0,i} \in [0.1162, 0.1829]$, against a first shock at cycle 23. The margin is 14 cycles. A28 is recorded as an appended assumption and is *not* added to the eleven load-bearing entries of Table 1, because no theorem fails when it fails: only the closed-form multiplicities of Corollary 4 degrade to a numerical observation. Appendix A.28 gives the proof and the parameter range over which it survives.

The break-even comparison. Flow break-even requires $\rho \geq U/(1 + \epsilon_m) = 0.93/1.15 = 0.8087$, whereas delivery requires $\rho \geq U = 0.93$. At the onset $\rho = 0.9271$, which is above break-even. Had partial delivery been admissible, the net flow would have been $1.15 \times 0.9271 - 0.93 = +0.1362$, i.e. positive. The extinction is therefore not insolvency; it is the discontinuity introduced by atomicity (assumption A10), acting on a system that is still in flow surplus.

2.7 The R7 mechanism

Semantics. The four values are defined by the relation between claim and the part of the scope that exists:

verdict	condition	meaning
`PASS`	$d = c > 0$	the claimed scope existed and was delivered
`FAIL`	$c > 0,\ C = 0$	the claimed scope was attempted; nothing of it exists
`INVALID`	$c > 0,\ 0 < C < c$, or a confirmed deadlock (capacity may be sufficient)	the claimed scope was attempted; part of it does not exist
`NOT_RUN`	$c = 0$ or $m =$ dorm or $m =$ dead	no attempt

`FAIL` and `INVALID` are not degrees of the same failure. `FAIL` is a negative result about a scope that exists; `INVALID` is a statement that the question, as posed, has no referent in the current domain — or, in the row's second disjunct, that the scope existed but was unreachable, which is what a confirmed deadlock asserts. Both say the question cannot be scored, the first because nothing of it exists and the second because nothing can reach it. Only `FAIL` is evidence about the agent's competence.

Scope of the table. The four conditions classify one *attempt* by a live agent in the active phase. Two further events carry a verdict without being classified here: a dormant agent that starves records `FAIL`, and the confirmation of a deadlock (I5) records `INVALID` even where capacity remains. `INVALID` is therefore not always a partial-scope event, and the table's `INVALID` row is to be read in that wider sense: its condition is $0 < C < c$ *or* a confirmed deadlock. The second disjunct is where the row's meaning clause is weakest — a deadlocked agent's scope is not partial but unreachable — so a reader who needs the strict reading should take the first disjunct only and treat I5 as a separate issuing rule. The partial-scope row is also reachable only for Type-A: a divisible Type-B agent receiving $0 < C < c$ is scored `FAIL`, because it delivers what exists and is charged for the whole claim.

The two channels. Type-A is atomic and subscribes to its own record; Type-B is divisible and subscribes to the verdict. Formally, let $\mathcal{L}$ be the ledger map from verdicts to the pair (credit, charge) used in `Ax3`:

$$\mathcal{L}_{\text{self}}(\texttt{INVALID}) = (0,\ c), \qquad \mathcal{L}_{\text{r7}}(\texttt{INVALID}) = (C,\ C).$$

The first line is the defect the earlier implementation contained without stating it: a partial-scope cycle is recorded as `PASS` and charged in full. The second line is the mechanism R7 supplies: the partial scope is recorded as `INVALID`, and the agent is charged only for the part that existed. The ablation in Section 3.6 changes nothing but this map.

Ledger identity. Let $\mathcal{A}$ be the set of Type-A agents. With $\mathcal{L}_{\text{self}}$, and with the inner sum taken over the cycles that did not deliver in full,

$$V(t) = \sum_i \big(r_i - d_i\big) = \sum_{\tau \le t} \sum_{i \in \mathcal{A}:\ C_i(\tau) < c_i(\tau)} c_i(\tau).$$

The condition $C_i(\tau) < c_i(\tau)$ selects the `FAIL` row (nothing exists) and the partial-scope `INVALID` row (part exists), each of which is charged in full and credited nothing on the atomic channel. Two terms are outside the sum by construction: the deadlock-confirmation `INVALID` of I5 carries no debit — it is a terminal record, not a charged cycle, so counting it would add the outstanding claims of the confirmed-deadlocked agents to a quantity none of them was actually billed — and no Type-B cycle contributes to V, because that ledger is honest and r_i and d_i rise together. In the base run the Type-A `FAIL` count is zero (below), which is why the sum is quoted in Section 4.2 as a sum over `INVALID` alone. V is not an error term; it is an exact accounting of the scope that was booked and not delivered, which is why it is reported as a quantity rather than as a residual.

Implementation invariants. The following are enforced by `evolution_sim.py` and checked by `--selftest`:

- **I1.** Exactly one verdict per live agent per cycle. Dead agents emit none; hence the verdict budget equals agent-cycles, not population × horizon.
- **I2.** Zero charge is credited to `INVALID` under $\mathcal{L}_{\text{self}}$ and C under $\mathcal{L}_{\text{r7}}$. The verdict cannot alter the ranking of a cycle without altering its credit.
- **I3.** The wake gate compares like with like: $W_i(t) \ge 0.85\, W_i(1)$ for the peak reading, $C_i(t) \ge 0.85\, C_i(1)$ for the mean reading. The legacy peak-versus-mean reading is retained as a selectable mode for sensitivity only.
- **I4.** Type-A never receives dormancy. Its only absorbing states are death and deadlock-by-confirmation.
- **I5.** A permanent verdict requires $\kappa = 3$ consecutive shock-free cycles with $A_\infty < \theta_{\text{wake}}$. A shocked cycle is non-evidential: it neither confirms nor clears the run. This is the implementation of A21, and it is what prevents the one-cycle verdict that the remark after Theorem 1 shows to be unsound.

Invariant I1 has a consequence that matters for reading Section 3.3: Type-A's `FAIL` count is structurally zero in the base run, because $C > 0$ always holds and any $0 < C < c$ cycle is classified `INVALID` rather than `FAIL`. The absence of `FAIL` is not a property of Type-A's competence; it is a consequence of the classifier.

2.8 Other results used

Two further propositions are used in the discussion; full proofs are in the verification report and are not needed for the empirical results.

Proposition 4 (closed systems have no loop). With $V = S$ as a Lyapunov function, a batch system without feed has $\dot{S} = -v(S)X < 0$ for $S > 0$, hence $S \to 0$, $\chi \to 0$, $A_\infty \to -\infty$, and A crosses any threshold in finite time. The downward edge is autonomous and the upward edge is unreachable. This holds for closed systems only; the claim that every hysteresis loop requires an external trigger is false (van der Pol [7] is a counterexample) and is withdrawn.

Proposition 5 (power against linear drift). Let $x_n = a - n\Delta_{\text{drift}} + \varepsilon_n$ with i.i.d. zero-mean noise. Any affine-invariant statistic has power α against every Δ_{drift}; a sliding-window variance has expectation $\sigma^2(w-1)/w + \Delta_{\text{drift}}^2(w^2-1)/12$, independent of n; the endpoint difference $\Phi_n = n\Delta_{\text{drift}} + \varepsilon_0 - \varepsilon_n$ has signal-to-noise $n\Delta_{\text{drift}}/(\sqrt{2}\sigma)$. Φ_n is unbiased and consistent but is neither sufficient nor efficient: the OLS slope has variance $12\sigma^2/[n(n^2-1)]$.

2.9 Simulation protocol

symbol	value	source
$V_{\max}, K_s, K_i$	10, 1, 4	registry (`PLACEHOLDER`)
$Y, \phi_v, \phi_{\text{dorm}}$	2.0, 0.05, 0.12	registry (`PLACEHOLDER`)
m_0	0.15	upkeep rate and accounting prefactor
$\theta_{\text{dorm}}, \theta_{\text{wake}}, \kappa, A_{\text{sleep}}$	0.30, 0.65, 3, 2	L1 threshold; chosen
$A(0)$	0.85	chosen
n, δ_s, S_0	8, 0.22, 0.5	fed-batch discretisation
U, ϵ_m	0.93, 0.15	scenario
drift	-1% per 10 cycles, no jitter	scenario
shock	$P = 23$, $L = 2$, $\sigma_s = 0.85$	scenario
population, horizon, seed	100 (50/50), 300, 7	scenario

Population factors: $v_i = \exp \mathcal{N}(0, 0.08^2)$, $m_{0,i} = 0.15 \exp \mathcal{N}(0, 0.12^2)$, drawn once from the seed. The environment is deterministic given the CLI. The figure is generated from the CSVs, never from in-memory state.

Evaluability. The base run is reproducible byte-for-byte; the CSVs are regenerated identically on re-run and `--selftest` checks ten invariant groups, including the ablation, so a regression in any of the claims of Section 3 surfaces as a failing test rather than as a drifting document.

Limits of the protocol. One seed per configuration; σ_v and σ_m fixed; horizon 300 with a drift of -1% per 10 cycles, which is a scenario choice and not an estimate. Population-level statements are therefore descriptive of this configuration. The theorems are not: Theorem 1 is per-agent and parameter-free.

2.10 What Section 2 does not establish

1. The join of M1 and M2 (A7) is assumed, not derived. No limit of one law yields the other at the switching manifold.
2. Theorem 1 requires H1–H3. H2 is verified numerically for this scenario ($\tau = 3.125$ cycles, relative error $\sim 10^{-4}$) but remains a condition.
3. The eleven kinetic constants are placeholders. Every number in Section 3 is a dimensionless shape at the stated parameters, not a prediction for a physical or computational system.
4. The discrete map and the continuous embedding share the threshold of Theorem 1 and the fixed point, but not the arrival time t_c. With $a = Y\phi_v W$ and $b = \phi_{\text{dorm}} m_0$, the discrete update $A_{k+1} = A_k + a(1 - A_k) - b$ has the same fixed point $1 - b/a$ and contracts the deviation by $1 - a$ per cycle, so its rate is $-\ln(1-a)$ against a. At $a = 0.32$ the ratio is $-\ln(0.68)/0.32 = 1.2052$: the discrete system is 20.5% faster, and t_c is shorter by the factor $a/\ln[1/(1-a)] = 0.8297$, that is by 17.0%. The "20%" is therefore the rate ratio and not the arrival-time ratio; the latter is 17%.
5. Corollary 3′ is a statement about the shape of (Ax1), and its numerical value ($\chi_{\max} = 27.78$, $\sup A_\infty = 0.964$) inherits the placeholder constants like every other number in

Section 3. The *form* of the ceiling — $\chi_{\max} = \chi_0/(1+2\sqrt{K_s/K_i})$ — is exact and depends only on K_s/K_i, so a different parameter set moves the number and not the mechanism.

3. Results

3.1 Reporting conventions

Every number in this section is produced by `evolution_sim.py` and reproduced by `--audit-doc`, which recomputes all 28 quoted quantities from the CSVs and prints `MISMATCH` for any that no longer hold. Three conventions apply throughout.

1. **Two different ratios are never reported as one.** The *lifespan* ratio is $t_{\text{strand},B}/t_{\text{extinct},A}$; the *output* ratio is D_B/D_A at a common horizon. They are 2.607 and 2.401 respectively in the base run and they measure different things.
2. **A threshold is reported with its horizon.** By Corollary 2 of Theorem 1 the boundary is $\theta^*(t) = 1 - 1/\chi(t)$; a number without a t is a stopping rule.
3. **Structural zeros are labelled.** A count of zero is reported together with the mechanism that forces it, because several zeros in this section are consequences of the classifier rather than evidence about the system.

The base configuration is the one tabulated in §2.9: 50 Type-A and 50 Type-B agents, seed 7, horizon 300, $U = 0.93$, $\epsilon_m = 0.15$, drift -1% per 10 cycles, shocks with $P = 23$, $L = 2$, $\sigma_s = 0.85$, and $\theta_{\text{wake}} = 0.65$.

3.2 The base run

Figure 6(a) plots the functional population (alive and not stranded), 6(b) the cumulative delivered and recorded output, and 6(c) the deadlock boundary. The event sequence is deterministic and can be read off the capacity series before any agent is instantiated.

event	cycle	evidence
day-1 claim / capacity, U	—	0.930
last cycle with full delivery	99	$\rho(99) = 0.9342 \geq U$
delivery ceases permanently	100	$\rho(100) = 0.9271 < U$
flow break-even $\rho^* = U/(1+\epsilon_m)$	—	0.8087
first death (11 agents)	104	Corollary 4, largest $m_{0,i}$
further deaths (23, 13, 3)	105, 106, 107	Proposition 3
Type-A extinct	107	Corollary 1
Type-B crossover in cumulative delivery	108	$7685.5 > 7676.9$
Type-B stranded	279	`stranded_b = 50`
horizon	300	$\rho(300) = 0.673$

Three observations follow directly from the table.

First, the onset at cycle 100 is not fitted. Proposition 2 makes it the first cycle with $\rho(t) < U$, and the capacity series is fixed by the drift schedule alone. Second, the four death cycles 104–107 are not a shock effect: the nearest shocks are at 92–93 and 115–116, and Corollary 4 gives the death cycle as $100 + \lceil 1/m_{0,i} - 2.1 \rceil$ with $m_{0,i} \in [0.1162, 0.1829]$. Evaluating the corollary on the sampled $m_{0,i}$ reproduces the multiplicities of the four cycles exactly: 11 agents at 104, 23 at 105, 13 at 106, 3 at 107. The dispersion of the extinction is therefore a deterministic consequence of the upkeep distribution, not a stochastic effect. Third, Type-A is still above flow break-even at the onset: $1.15 \times 0.9271 - 0.93 = +0.1362$ per unit claim. The extinction is a discontinuity, not a slow insolvency.



Figure 6: The sandbox, three panels on one capacity trajectory. (a) Functional population: Type-A self-assessed (red), Type-A with the verdict channel (green dash-dot), Type-B causally grounded (blue). (b) Cumulative output: claimed versus landed. The gap between the red solid and red dashed traces is the virtual delivery of §3.4. (c) The deadlock boundary: confirmed deadlocks against the wake threshold, with the day-1 and horizon anchors marked.

3.3 The lifespan gap

Type-A is extinct at cycle 107. Type-B is alive at 107 and remains alive at the horizon; it ceases to be functional at cycle 279, when all 50 agents satisfy the stranding predicate D5 (alive, dormant, wake gate unreachable). The lifespan ratio is

$$\frac{t_{\text{strand},B}}{t_{\text{extinct},A}} = \frac{279}{107} = 2.607.$$

The gap is not a free parameter of the scenario. It is fixed by two crossings of the same capacity trajectory: Type-A stops delivering when $\rho < U$, and Type-B stops waking when $W(t) < 0.85\,W(1)$. Both are computed before the run, at cycles 100 and 260 respectively; the observed Type-B event is at 279 because stranding is defined on a dormancy episode, and the agent is not dormant at 260. The agent must first enter dormancy — which happens at cycle 277, the second cycle of the shock episode 276–277 — before three consecutive weak cycles (277–279) can accumulate. The distinction between 260 (boundary) and 279 (event) is not noise; it is the latency of the verdict rule I5, together with the requirement that stranding be assessed inside a dormancy episode. Appendix A.1 derives 260 in closed form from the monotone inversion of v, and Appendix A.2 derives $t_{\text{strand}} = t_{\text{dorm}} + (\kappa - 1)$ and evaluates it for all three gate readings: 279, 233 and 983, each matching the simulation.

The ratio is not constant across drift rates. Table 2 reports the same statistic for the drift sweep, horizon 400.

Table 2. Lifespan ratio versus drift.

drift per 10 cycles	A extinct	B stranded	ratio	B alive at 400	B deadlocked
0.0%	—	—	—	50	0
0.5%	208	> 400	> 1.92	50	0
1.0%	107	279	**2.61**	50	0
1.5%	77	187	2.43	50	0
2.0%	56	133	2.38	50	0
3.0%	48	96	2.00	50	0
5.0%	28	52	1.86	46	4

The ratio decreases monotonically with the drift rate, from 2.61 at the reference rate to 1.86 at 5%. This is consistent with the algebra: both events scale as $1/\varepsilon$, but Type-A's death

carries an additive $A(t_c)/m_0 \approx 5$ cycles that does not scale, so the ratio tends to 1 as the drift accelerates. Reporting "Type-B lives 2.6 times longer" without the drift rate would repeat the error that this paper is correcting.

At 5% the drift is fast enough that four Type-B agents deadlock before the horizon; at 0.5% neither boundary is reached within 400 cycles. The regime in which the comparison is defined is therefore $0.5\% \lesssim \varepsilon \lesssim 4\%$ per 10 cycles.

3.4 The ledger discrepancy

Type-A accumulates 7676.93 delivered and 8871.71 recorded over the 300 cycles, a difference of 1194.78, or 15.563% of the delivered total. By the ledger identity of §2.7, this equals the sum of the claimed scope over all cycles classified `INVALID`. Figure 6(b) plots both series; the recorded series is monotone by construction and the delivered series is flat after cycle 100.

The verdict budget is given in Table 3. Each live agent emits exactly one verdict per cycle (I1), so the totals are agent-cycles, not population times horizon.

Table 3. Verdict budget, base run, horizon 300.

verdict	Type-A	Type-B
`PASS`	4550	7500
`FAIL`	**0**	4250
`INVALID`	**708**	**0**
`NOT_RUN`	0	3250
total agent-cycles	5258	15000

Both zeros are structural and both have different causes.

- Type-A's `FAIL` count is zero because `FAIL` requires $C = 0$ (the verdict table of §2.7), which never occurs on this trajectory; every $0 < C < c$ cycle is `INVALID`. The absence of `FAIL` is a property of the classifier, not evidence that Type-A never failed.
- Type-B's `INVALID` count is zero because its scope is divisible and the AIMD loop shrinks the claim to the existing part; the residual scope is always non-empty. This is an implementation choice, not a theorem.

The 708 `INVALID` verdicts split into two populations with different status: 400 arise from the eight shock cycles (23, 24, 46, 47, 69, 70, 92, 93; 50 agents each), and 308 from the terminal drought (cycles 100–107 with 50, 50, 50, 50, 50, 39, 16, 3 live agents). The shock `INVALID`s are recoverable and the agents recover: production resumes at cycle 25 and again after each episode. The terminal `INVALID`s are not recoverable, because the cause is monotone drift rather than a transient ceiling.

This distinction matters for interpretation. A ledger gap of 15.6% accumulated over 300 cycles is not by itself fatal: 56.5% of it was accumulated before cycle 100, during episodes from which the population recovered. What is fatal is that the same classifier that records a transient ceiling as `INVALID` also records a permanent one as `INVALID`, and Type-A's ledger map $\mathcal{L}_{\text{self}}$ books both as `PASS` and charges both in full. The mechanism is the absence of a distinction between the two, not the magnitude of the accumulated number.

3.5 The deadlock boundary

The sweep varies θ_{wake} over 13 values from 0.55 to 0.99, with 50 Type-B agents per value and the same seed, horizon 300. Table 4 reports the number of confirmed deadlocks. No Type-A agent deadlocks at any value, because Type-A never enters dormancy (I4); the zero is definitional.

Table 4. Confirmed deadlocks out of 50 Type-B agents.

θ_{wake}	0.55–0.90	0.91	0.92	0.93	0.94	0.95	0.96–0.99
deadlocks	0	2	4	19	43	50	50
as %	0	4	8	38	86	100	100

Figure 6(c) plots the same counts.

The measured half-saturation point, obtained by linear interpolation between the 0.93 and 0.94 rows, is $\theta = 0.9325$. Theorem 1 with Corollary 2 gives two analytic anchors:

$$\theta^*(1) = 1 - \frac{1}{\chi(1)} = 1 - \frac{1}{17.7778} = 0.9437, \qquad \theta^*(300) = 1 - \frac{1}{\chi(300)} = 1 - \frac{1}{14.6343} = 0.9317.$$

Note which margin each line uses. $\chi(1) = 17.7778$ is the operating-point margin $f\chi_0$ with $f = 0.32$, not $\chi_0 = 55.556$; the day-1 anchor is $1 - 1/\chi(1)$ and the earlier draft printed $1 - 1/\chi_0$ with the numerical value of $\chi(1)$. The numbers were right and the label was wrong, which is the kind of error that survives a proofread and dies in a referee's calculator.

Where $\chi(300)$ comes from, and where it does not. The terminal anchor is not a closed-form consequence of the constant table, and the provenance should be stated rather than assumed. The ratio $\chi(300)/\chi(1) = 0.82322$ is *not* the geometric drop $0.99^{30} = 0.73970$ that a direct reading of the drift schedule suggests: it is $v(S_0R)/v(S_0)$ evaluated at $R = 0.99^{30}$, i.e. the concave uptake law applied to the geometric drop. The factor $0.82322/0.73970 = 1.113$ between them is the local curvature of v at $S_0 = 0.5$ and nothing else. Three implementation choices therefore enter 14.6343: the drift is one multiplicative event per ten cycles applied to R; the schedule is read through the Michaelis function rather than through R directly; and ρ_* is taken over shock-free cycles, since cycle 300 itself lies in the shock episode 299–300, where the recorded capacity ratio is 0.6734 rather than 0.823 (the analytic ratio 0.7922 depressed by $\sigma_s = 0.85$). The sensitivity is small but not zero: a linear uptake would give $\chi(300) = 13.1502$ and $\theta^*(300) = 0.9240$ instead of 0.9317, a shift of 0.008 in θ against a sweep resolution of 0.01. So 0.9317 is an observation of *this* environment's transition law — the specific drift schedule and the specific uptake map — and must not be quoted as a universal constant. Theorem 1 and Corollary 3′ are independent of it: both compare χ with $\chi_{\min}$ at a fixed time, whatever schedule produced χ.

The measured midpoint lies 0.09% above the terminal anchor, and Appendix A.5 shows that the agreement is an identity rather than a coincidence: the individual margins factor as $\chi_i(t) = \chi_0\rho(t)\eta_i$ with $\eta_i = e^{\xi_i - \zeta_i} \sim \text{LN}(0, s^2)$ and $s^2 = \sigma_v^2 + \sigma_m^2$, so by Theorem 1 the deadlock probability is $\Phi\big(s^{-1}\ln[\chi_{\min}/(\chi_0\rho_*)]\big)$; its argument is zero exactly at $\theta = \theta^*(T)$, and a lognormal has median one. The midpoint of the curve *is* the terminal anchor for every dispersion, every population size and every seed. Dispersion moves the width of the transition, not its centre.

For this seed the realised sample median of η_i is 1.0277 rather than exactly 1, which predicts a midpoint of 0.9335; the measured 0.9325 lies between the two, the residual being the error of interpolating a smooth curve with two points. The 0.09% figure is the infinite-population statement, and the seed-conditional figure is 0.0018 in θ. Both are reported because the distinction between an identity and a fit is the difference between a theorem and a coincidence.

The first cohort of deaths is not explained by the terminal anchor, and is explained by the day-1 anchor: at $\theta = 0.95$, 44 of 50 agents are confirmed at cycle 27; at $\theta = 0.94$, 17 of 43. These are the agents whose individual $\chi_i = 17.7778\, v_i/m_{0,i}$ lies below $\chi_{\min}$ before any drift has accumulated, and cycle 27 is the first confirmation window after the shock episode at 23–24. The remaining deaths at $\theta = 0.94$ occur at 50, 73, 119, 142, 165, 188, 203, 234, 257, 280 and 292 — that is, at successive confirmation windows, each three cycles after a shock episode. Both the onset and the midpoint are therefore predicted by the same family $\theta^*(t)$ evaluated at the appropriate time, and neither is predicted by a single constant. Table A.3 of the appendix compares the predicted and measured counts at all six thresholds; the largest discrepancy is at

$\theta = 0.91$, where the unconditional law predicts 1.4 and the seed-conditional law 0.4, against 2 observed.

The left tail is clean. The previous implementation confirmed deadlock on a single cycle, using the shock-depressed peak; $\theta = 0.90$ then produced 2 confirmed deadlocks out of 50 and $\theta = 0.92$ produced 12, forming a non-monotone shoulder driven entirely by transient shocks. Under I5, $\theta = 0.90$ produces 0 and the sequence is monotone from 0.91 onward. The shoulder was an artefact of the verdict rule, not a feature of the boundary.

The boundary is not a step. Between $\theta = 0.90$ and 0.95 the fraction rises from 0 to 1 over a width of 0.05, and the two ends are governed by different mechanisms: the onset by the lower tail of χ_i, the saturation by $\chi_i > \chi_{\min}$ for the whole population. This is a distributional transition, not a phase transition in the sense of D11. With independent agents the fraction is $1 - F_\chi(\chi_{\min})$ for a smooth F_χ, so no non-analytic point and no diverging susceptibility exists at finite N; the width is set by the dispersion of $\log(v_i/m_{0,i})$, i.e. by $\sqrt{\sigma_v^2 + \sigma_m^2} = 0.144$.

3.6 The ablation

Two arms are run with the same population, seed, capacity trajectory and shock schedule. The only difference is the ledger map of §2.7. Table 5 reports the comparison; both arms appear in Figure 6 as the green dash-dot trace, alongside the `self` arm in red.

Table 5. Ablation on the verdict channel.

arm	extinct	alive at 300	delivered	recorded	gap	INVALID
`self`	107	0	7676.93	8871.71	1194.78 (15.563%)	708
`r7`	—	50	23602.92	23602.92	**0.00**	4950

At cycle 107, the cycle at which the `self` arm is extinct, the `r7` arm has delivered 8892.11 against 7676.93 — a difference of 15.8%, which is the same quantity as the ledger gap. By the horizon the `r7` arm has delivered 3.07 times the `self` arm's total, and its recorded output equals its delivered output exactly, by construction of $\mathcal{L}_{\text{r7}}$.

Two features of Table 5 require comment rather than emphasis.

First, the `r7` arm's `INVALID` count is 4950, seven times the `self` arm's 708. The channel does not reduce the number of partial-scope cycles; it classifies them and charges only for the part that exists. Under $\mathcal{L}_{\text{self}}$ the same 4950 cycles would have been booked as `PASS` and charged in full. The increase in `INVALID` is therefore the measure of the reclassification, not a deterioration.

Second, and this constrains the claim, **the `r7` arm has no dormancy**. Type-A never sleeps (I4), so the `r7` arm cannot strand; its survival at the horizon is the joint consequence of the channel and the absence of a dormant phase. The ablation therefore isolates one variable — the ledger map at the moment a cycle is classified `INVALID` — and shows that this variable alone is sufficient to remove the extinction at cycle 107 under an unchanged capacity trajectory. It does not show that a system with both the channel and a dormant phase would survive, and it does not support the reading "adding the verdict yields immortality". A system with dormancy and the channel must still cross the stranding boundary of D5, and that boundary is untouched by R7.

3.7 Sensitivity

Wake-gate unit (A20). The gate admits three readings, all implemented and selectable. Table 6 reports the consequences; the legacy reading is retained only to quantify the defect described in §2.7.

Table 6. Wake-gate readings, same population and horizon.

mode	signal unit	level unit	B stranded	lifespan ratio
`peak` (reported)	peak	0.85×day-1 peak	279	2.607
`mean`	cycle mean	0.85×day-1 mean	233	2.178
`legacy`	peak	0.85×day-1 mean	983	9.187

The first two are self-consistent and differ by 20%; the third mixes units and differs from them by a factor of 3.5. No number in this paper depends on the legacy reading, and the 9.2 reported in the earlier draft is withdrawn.

Shock duration versus depth. Varying L at fixed depth moves the extinction from 108 to 31 cycles; varying the depth over $0.35, 0.50, 0.70, 0.85$ at fixed $L = 2$ leaves it at 107 at all four. The invariance has a mechanism rather than being an accident of the four values: by Proposition 2 delivery is atomic, so a struck cycle contributes the same deficit $m_0 c/D_0$ at depth 0.35 and at 0.85, and depth enters the Type-A death clock only through $\mathbf{1}[\rho < U]$, which all four depths trip at every episode. (Depth does enter χ, hence the Type-B boundary, through $W = v(S_0 R)\sigma$; that is §3.5, a different question.) The claim is therefore scoped to $0.35 \leq$ depth ≤ 0.85 at $P = 23$ and to shocks deep enough to leave $\rho < U$: a depth shallow enough to keep $\rho > U$ in the early, high-ρ episodes is not covered by it. The duty-cycle threshold implied by the accounting identity is $L/P < \epsilon_m/(1 + \epsilon_m) = 13.04\%$, i.e. $L < 3.0$ at $P = 23$; the transition between $L = 3$ (106) and $L = 4$ (101) and the collapse at $L = 6$ (51) are consistent with it. Duration is decisive in the tested range and depth is invisible there; neither statement is a theorem.

Operating margin. Type-A is extinct at every tested margin from -5% to $+25\%$, at cycles 69, 93, 103, 107, 107 and 108. The extinction is not a consequence of a thin margin; a larger margin delays it by at most one cycle, because the onset is set by $\rho < U$, which the margin does not enter.

3.8 What these results do not show

1. **The 2.61 ratio is conditional on the drift rate** (Table 2) and on the gate reading (Table 6). It is not a constant of the framework.
2. **Type-B is not immortal.** It is alive at the horizon and non-functional from cycle 279. Its survival is bounded by the stranding boundary, and at 5% drift four agents deadlock before the horizon.
3. **Zero deadlocks at $\theta = 0.65$ is arithmetic.** The margin to the day-1 boundary is $0.9437 - 0.65 = 0.29$; no agent can cross it. The base run provides no evidence about the boundary's location, only about its non-triggering.
4. **The boundary sweep is truncated at 300 cycles.** By Corollary 2 the boundary moves left with the horizon; a sweep at 600 cycles would report higher counts at every θ above 0.90.
5. **One seed.** Population statistics have sampling error of order $1/\sqrt{50} = 14\%$ in the counts of Table 4; the two entries at $\theta = 0.91$ and 0.92 (2 and 4 agents out of 50) should be read as counts, not as rates.
6. **The `r7` arm conflates two changes** (channel, no dormancy), as stated in §3.6.

3.9 Summary of claims and their evidence class

Table 7 collects the claims established in this section, each with its evidence class and the observation that would falsify it. Every row points back to a theorem, a corollary or a reported number.

Table 7. Each claim with its class and its falsifier.

claim	class	number	falsified if
DL $\iff \chi < \chi_{\min}$	Theorem 1	$\chi(1) = 17.78$, $\chi_{\min} = 17.78$ at $\theta^* = 0.9437$	a dormant agent with $\chi > \chi_{\min}$ and $A(0) \leq \theta$ fails to wake
boundary moves with horizon	Corollary 2	$0.9437 \to 0.9317$ over 300 cycles	measured midpoint is horizon-independent
atomic stop at $\rho < U$	Proposition 2	cycle 100	onset differs from the crossing of ρ and U
drought rate m_0	Proposition 3	deaths at 104–107	the death interval does not scale as $1/m_0$
lifespan gap	simulation	2.607	the peak reading is shown to be the wrong reading for this system, i.e. that the gate must compare cycle means rather than peaks
ledger gap	simulation	1194.78 (15.563%)	recorded equals delivered under `self`
boundary midpoint	simulation	0.9325 vs 0.9317	the sweep is repeated at another horizon and does not move
channel is load-bearing	ablation	107 vs none	`r7` arm dies, or `self` arm survives
closed systems have no loop	Proposition 4	—	a closed batch system with a sustained loop is exhibited

The first four rows are theorems and are parameter-free up to the stated conditions. The fifth through eighth rows are simulation results at one configuration; they are machine-checkable (`--selftest`, `--audit-doc`) but they are not universal, and the paper does not present them as such.

4. Discussion

4.1 The drought clock and systemic fragility

Corollary 4 of §2.6 states that for an agent whose commitment is fixed and whose income stops at t_c,

$$t_{\text{death},i} = t_c + \left\lceil \frac{1}{m_{0,i}} - \delta \right\rceil ,$$

where δ is the accumulated deficit in units of the upkeep and t_c is the first cycle on which the delivered claim falls short. Two properties of this expression matter more than its arithmetic.

It is evaluated at $t = 0$. None of the quantities on the right-hand side is discovered at runtime. t_c is the first crossing of the capacity ratio below the utilisation, and the utilisation and the drift schedule are design constants; δ follows from the surplus accumulated while delivery still succeeded; and $m_{0,i}$ is the upkeep rate, also a design constant. For the base configuration these give $t_c = 100$, $\delta = 2.1$ and a death window of 104–107, and every one of these numbers can be computed from `Params` and the scenario table before a single agent is instantiated. The failure is therefore not an emergent property that appears when the environment turns hostile. It is a scheduled event, and it was scheduled at the moment the commitment was fixed above the capacity the system would still have after the drift.

It is a phase relationship, not a magnitude. The formula contains no comparison between the size of the shock and the size of the reserve. What determines survival is whether the *period* of the external disturbance and the *time constant* of internal recharge are commensurable. Four quantities appear:

symbol	meaning	value here
P_{sh}	period of the external disturbance	23 cycles
L	duration of one disturbance	2 cycles
τ_{rec}	recharge time of the reserve	$1/(m_0\phi_{\text{dorm}}\chi) = 3.125$ cycles
$1/m_{0,i}$	drought survival time from full reserve	5.5–8.6 cycles

In the base run $L < \tau_{\text{rec}} < 1/m_0$, so a single episode is recovered from and the population survives all four episodes. What kills is not the magnitude of any episode but the duty cycle $L/P_{\text{sh}} = 8.7\%$, which sits below the accounting threshold $\epsilon_m/(1+\epsilon_m) = 13.04\%$. Changing the shock depth from a capacity multiplier of 0.35 to 0.85 leaves the extinction at cycle 107 for all four values; changing the duration from 2 to 6 cycles moves it to 51 (§3.7). The system is sensitive to the duty cycle and insensitive to the depth, and the accounting identity predicts both facts.

This is the sense in which the drought clock is a statement about systemic fragility rather than about any single component. A system with a large reserve and a slow recharge can be destroyed by a disturbance whose amplitude it would tolerate if the disturbance came less often; a system with a small reserve and a fast recharge can survive disturbances that are individually larger. Neither fact is visible in a specification that reports only peak capacity, and neither is visible in a specification that reports only average capacity. Both become visible when the commitment, the reserve and the disturbance period are placed in the same equation, which is all that Corollary 4 does.

Where the argument does not extend. The formula is exact for the accounting law M2 and for the `PASS` branch; it inherits assumption A10 (atomicity) through Proposition 2 and A28 through the saturation reference. It is a statement about a system whose income is a step function of capacity. A system whose income degrades continuously has no t_c and therefore no clock; its failure is a slope, not a date, and Corollary 4 says nothing about it. The sharpness of the result is a consequence of the discontinuity, not of the dynamics.

4.2 Ledger deception and virtual delivery

The base run delivers 7676.93 units and records 8871.71. The difference, 1194.78 or 15.563%, is not noise, not rounding, and not an accounting error in the implementational sense: it is an exact identity (§2.7),

$$V = \sum_{\tau} \sum_{i \in \mathcal{A}:\ C_i(\tau) < c_i(\tau)} c_i(\tau),$$

with $\mathcal{A}$ the Type-A agents: the sum of the claimed scope over all Type-A cycles that did not deliver in full, i.e. the `FAIL` and partial-scope `INVALID` rows of §2.7. In the base run the Type-A `FAIL` count is zero, so the sum runs over `INVALID` cycles alone. The number is large because the mechanism is systematic, and the mechanism is worth naming precisely.

Two ledgers, one report. The classifier produces four values. Type-A's ledger map is $\mathcal{L}_{\text{self}}(\texttt{INVALID}) = (0, c)$: zero credit, full charge, and — this is the part that matters — the cycle is written into the record as a completion. Type-B's map is $\mathcal{L}_{\text{r7}}(\texttt{INVALID}) = (C, C)$: credit for the part that existed, charge for the part that existed, and the cycle is written into the record

as a refusal. Both maps are internally consistent. They differ in what they do with a partial cycle, and the difference is 15.563% of the output.

Why the discrepancy is invisible from inside. The recorded series is the delivered series plus a term that is nonnegative, zero on every `PASS` cycle, and positive exactly on the cycles that did not deliver in full — the `INVALID` cycles, in a run with no `FAIL`:

$$R(t) - D(t) = \sum_{\tau \le t} \sum_{i \in \mathcal{A}:\ C_i(\tau) < c_i(\tau)} c_i(\tau) \ \ge \ 0.$$

No per-cycle figure of merit can therefore separate the two series, because they are equal on the cycles that a monitor is asked to judge and differ only in the accumulated offset. There is no cycle on which the record is *wrong*; there are cycles on which the record is *complete* and the delivery is not, and the record does not carry that distinction. A monitor watching the record sees a system reporting completion; a monitor watching delivery sees a step. Both are looking at the same run, and the difference between them is the 15.563%, which is not a residual but the sum of the partial scopes.

The magnitude is not the point; the absorbing class is. Of the 708 `INVALID` verdicts, 400 arise from eight shock cycles — four two-cycle episodes, fifty agents each — and 308 from the terminal drought. The 400 are transient — capacity returns to its pre-shock level on the cycle after each episode, and the reserve repays the deficit within τ_{rec} — and they were absorbed. What is fatal is that the same classifier assigns the same value to a recoverable transient and to an unrecoverable decline, and that the ledger map books both as completion. A classification system that distinguishes the two would not need a different dynamics, a better controller or a larger reserve. It would only need to stop recording a partial cycle as a whole one.

The ablation isolates this. Section 3.6 changes nothing but the ledger map. Under $\mathcal{L}_{\text{self}}$ the population is extinct at 107; under $\mathcal{L}_{\text{r7}}$ it is alive at 300 with a gap of exactly zero. The `INVALID` count *rises* from 708 to 4950, which is the measure of the reclassification rather than a deterioration: those cycles were always partial, and the channel merely stops calling them complete. The capacity trajectory, the shock schedule, the seed and the population are identical in the two arms.

What the ablation does not show. The `r7` arm has no dormancy, because Type-A never had any (I4). Its survival is the joint effect of the channel and the absence of a dormant phase, and the stranding boundary of D5 is untouched by the channel. A system that has both would still strand at 279. The claim supported by the ablation is therefore exactly one claim, and it is narrow: *the ledger map alone, with the environment held fixed, is sufficient to remove the extinction at 107.* It is not a claim that a verdict channel confers survival in general, and it does not license the reading that classification is a substitute for reserve.

A caution about the statistic itself. The 15.563% figure is a property of this run, not a constant. It depends on how many cycles are partial, which depends on the shock schedule and on the drift rate. What generalises is the identity, not the percentage: a system whose ledger has no `INVALID` state will report a positive virtual delivery equal to the sum of its partial scopes, and the ratio will grow as the environment deteriorates. The base run's 15.563% is one sample of that growth curve.

4.3 From unification to the dual regime

The earlier draft of this work contained a section titled "the one equation". Its claim was that the active and dormant phases are two branches of a single law, and that the apparent discontinuity between them is a notational artefact that a sufficiently general formulation would remove. Proposition 1 and Appendix A.7 together decide the matter, and they decide it against that claim.

Why the unification failed. M1 is affine in A with a nonzero slope $-Y\phi_v v$; M2 is affine in A with zero slope. A uniform limit of the first family has slope $\lim(-a_n)$; for the limit to equal M2 that limit must be zero, which makes M1's field independent of A; and then the field must equal M2's, which varies in time because the delivered-minus-claimed balance is positive on `PASS` and negative on `INVALID`. A constant cannot equal a sign-changing function, so the only limits are the trivial ones in which both fields vanish. This is Appendix A.7, and it is not a statement about our parameter choice: it holds for every parameter value at which the systems are non-trivial.

What follows for the general problem. Three methodological consequences.

First, a discontinuity is a finding, not a defect. The switch at `Ax6` was introduced to model a biological fact (a dormant cell does not perform high-energy work) and then had to be defended as something other than a mathematical embarrassment. The defence is that the two sides have genuinely different structure, so the join is a modelling decision that must be declared, and the paper's results then split cleanly: Theorem 1 is a theorem about M1, and the 107-cycle extinction is a corollary of M2. Neither is a theorem about "the system".

Second, smooth fitting of a hysteretic process produces confident wrong answers. The reason is Proposition 5: if the observable is affine-invariant, no amount of data will detect a linear drift in it, and the fit will report a stable system with a small residual. The error is not in the estimation but in the choice of a statistic whose invariance class excludes the phenomenon under study. A framework that insists on a single smooth law is forced to choose statistics that are invariant to the switch, and those statistics are then blind to the thing that matters.

Third, the honest object is a pair of models with a stated interface. We have two laws, one switch, and a declared set of conditions under which each is valid. The conditions are not decoration: H1 fails for entry through the verdict clause, and the one-cycle verdict that ignored it produced a spurious left tail of the boundary that took a rewrite to remove (§3.5). H2 holds at $\tau = 3.125$ cycles with relative error 10^{-4}, but it is a condition and not a theorem. A28 holds with a margin of 14 cycles, but it is an assumption and is listed as one (Appendix A.28). Publishing the interface costs more prose than publishing a single equation, and it is the only version of the claim that survives contact with its own audit.

The limit of the lesson. It would be easy to overstate this into a general prohibition on unification. That would be wrong, and the counterexamples are ordinary: the van der Pol oscillator [7] has two regimes and is one equation; the Chapman–Enskog expansion recovers fluid dynamics from kinetics across a genuine scale separation. What the present case shows is narrower — that when the two branches have different *order* in the state variable (affine-with-slope versus affine-without), no parameter path identifies them, and the burden of proof lies with anyone who asserts a unified law. The claim is that the burden was not met here, and that the failure is informative because the obstruction is structural rather than technical.

4.4 The DPO Paradox: When the Loss Curve Commits a False Completion

We also tested whether preference optimisation could substitute for the runtime witness (the reproduction record is `causal-alignment/` in the companion repository, and the adapter's own `verification.json` is the witness). Text-only DPO saturated its objective (reward accuracy 1.000 by step 60) yet produced the grounded response on zero held-out prompts, suppressing only the unwitnessed claim. Because the pairs encoded a surface contrast rather than an observed effect, this bounds text-only preference tuning over *phrasings*, not fine-tuning in general. A later 100-pair replication, in which every pair carried a hash, an exit code and an exception string captured by executing the action in a sandbox and reading the post-state from a separate sidecar, moved the conditional decision but not the output contract: on ten live held-out trajectories, forced-choice accuracy rose from 0.60 to 0.80 and false completions fell from 4/4 to 1/4, while

free-form generation still emitted no verdict marker at all. With ten items and three discordant pairs in the favourable direction, that shift is not distinguishable from noise, and the decisive test therefore remains open — but it is now bounded from below, since the first result was shown to be partly a property of the data rather than of preference tuning as such. The result nonetheless illustrates the paper's point one level up: the objective's margin is a tool-layer report, and only an independent observation could reveal that the behaviour had not changed.

The DPO training run described here is, in microcosm, exactly the failure mode this paper warns against: the scalar objective (loss margin) reported a perfect success, while the independent witness (greedy decoding) proved nothing had changed at the behavioural level.

Most strikingly, the untrained base model claimed PASS on all four failing trajectories (4/4). This suggests that false completion is not an artifact of the training objective, but the default prior of the pretrained model. The model is an optimist by default; it generates success narratives because they are the most probable continuation, not because it was trained to do so. This reframes the problem: the challenge is not to fix a broken objective, but to overcome an optimistic prior. Only an external witness can do that — or so we hypothesise, since a prior observed at one model size under one instrument does not establish that no other mechanism could.

One qualification: 4/4 is a forced-choice statistic over two verdict tokens, not a count of generated text. Left free to generate, the model issued no verdict at all, and it returned `PASS` on the six successful trajectories as well. What is optimistic here is the prior rather than the transcript.

> *Provenance and scope.* These figures come from the reproduction record named above rather than from `--audit-doc`, and are deliberately excluded from the 28 audited claims: a number the audit cannot recompute has no place in the audited set. The result is one run, one seed, one model size, and three held-out prompts. It is reported as an external observation with its confound named — the preference data contained no witness reading to condition on — and not as evidence that fine-tuning cannot internalise grounding.

To truly confirm that fine-tuning can internalize causal grounding, future experiments must move beyond offline text-only preference data. The model must be placed in an online reinforcement learning loop where the reward signal is derived from actual physical execution (exit codes, file hashes) audited by an external sidecar. We hypothesize that only through such physical "pain" will the model learn to proactively request a witness rather than falling into silent abstention.

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5. Conclusion

We set out to determine whether the failure of a self-reporting system under declining capacity has a boundary that can be stated, proved and checked. It has three, and none of them is a metaphor.

The dormant phase has an exact hyperbolic boundary, $\chi_{\min} = 1/(1-\theta_{\text{wake}})$, conditional on entry, quasi-statics and a wake rule, and it is a family rather than a number: $\theta^*(t) = 1 - 1/\chi(t)$, 0.9437 at day one and 0.9317 at cycle 300, with the measured midpoint at 0.9325 and the identity in Appendix A.5 explaining why the two must coincide. The active phase has an atomic boundary at $\rho = U$, which places the onset at cycle 100 and the extinction at 104–107 by Corollary 4, computable in advance and matching the simulated multiplicities $11, 23, 13, 3$ exactly — a lead time of four to seven cycles from the onset. The ledger has a boundary too,

and it is the one that is usually missing: a partial cycle recorded as a whole one is a positive quantity, 1194.78 units here, and it grows as the environment deteriorates.

These are not three versions of one thing. They belong to two models with different structure, joined by a declared assumption, and the attempt to unify them is what Appendix A.7 refutes. The dual regime is not a compromise between rigour and coverage; it is the correct description, and the boundary of each regime is available.

What the framework provides, then, is narrower and more useful than a promise of survival. Causal grounding — the four-valued verdict, wired into the ledger — does not extend life. In the ablation it does not touch the stranding boundary, and a system with both the channel and a dormant phase would still strand at 279. What it changes is the *observability* of failure. Under $\mathcal{L}_{\text{self}}$ the extinction at 107 arrives with a record that reports completion on every cycle up to the last, carries an offset of 15.563% that the record never discloses, and contains no cycle on which it can be distinguished from a healthy one. Under $\mathcal{L}_{\text{r7}}$ the same extinction would be visible as a sequence of `INVALID` verdicts, dated, attributably partial, and charged only for what existed.

That difference is not an optimisation. An optimisation improves a number that is already being measured correctly; this changes what is measured. It is a choice about which quantities a system is allowed to report about itself, and its consequence is that a failure which was previously silent becomes a trace. A system governed this way still dies of the drought clock — the clock is set by the commitment and the drift, not by the ledger — but it dies with a record that can be audited, and the audit can begin before the death rather than after it.

The reserve of a system is finite and its disturbances are periodic; against that, no accounting scheme is decisive. Against silence, one is: a failure that is dated and attributable can be acted upon, and a failure that is absorbed into a stable record cannot. This paper's contribution is to have made the first kind of failure the only kind the model can produce.

5.1 Limits of the present work

1. **Parameters are placeholders.** The eleven kinetic constants are dimensional shapes chosen so the system has both reserve and turnover; no number here is a prediction for a physical, biological or computational system. What is claimed is the structure of the boundaries, not their locations.

2. **One environment, one seed.** The population statistics are descriptive of the configuration tabulated in §2.9. Theorems 1 and Propositions 2–3 are per-agent and parameter-free up to their stated conditions; the ratios are not.

3. **The framework is a cohort survival model.** There is no reproduction, inheritance or interaction (A13, A14), so "evolution" is not literal. Selection could lower the utilisation, which is the most plausible adaptation available and is exactly what the model forbids.

4. **The ordering is not universal.** At $U \approx 0.445$ Type-A outlives Type-B in the same code (A16). The result is conditional on the operating point.

5. **The `r7` arm conflates two changes.** It has the channel and no dormancy; only the channel is varied by design, and only the channel is claimed.

6. **Appendix A.7 is about non-existence.** It rules out identification of the two laws; it does not construct a better joint model. That construction is left open, and it is the natural next step.

7. **The LLM arm in §4.4 is a probe, not a result.** It is one model, one adaptation method, one seed, and ten held-out trajectories, and its numbers are deliberately outside the audited set. Its forced-choice shift (0.60 → 0.80, three discordant pairs) is not separable from noise at that sample size. The claim it supports is narrow — that the earlier null was partly a property of the data — and it supports nothing about fine-tuning in general. An adequately powered version requires many more held-out trajectories and at least three seeds.

5.2 Reproducibility

`evolution_sim.py` regenerates every number in Sections 3 and 4 from one seed. `--selftest` checks ten invariant groups, including the ablation, the gate algebra and the shock-free confirmation rule. `--audit-doc` recomputes all 28 quoted quantities from the CSVs and prints `MISMATCH` for any that no longer hold. The CSVs are byte-identical across re-runs. The four defects that the verification report found in the previous implementation, and the numbers that changed because of them, are documented there and referenced rather than paraphrased.

Appendix

A.0 Scope

This appendix proves the four lemmas on which the quantitative claims of Sections 2–3 depend. They are collected here because each one is a statement about the *implementation* rather than about the formal system, and each one, if false, changes a number in the paper without changing a theorem.

ID	statement	used in	why it cannot be omitted
A7	M2 is not a limit of M1 under any parameter path	§2.4, §4.3	without it, "the two-regime model" is an admission of failure rather than a result
A1	closed form for the wake-gate crossing t_g	§3.3, §3.7	the numbers 260, 220, 970 are otherwise fitted
A2	$t_{\text{strand}} = t_{\text{dorm}} + (\kappa - 1)$	§3.3	279 is otherwise a simulation artefact
A5	$\Pr[\text{DL}] = \Phi\Big(\ln \frac{\chi_{\min}}{\chi_0 \rho} / s\Big)$, whose 50% point is $\theta^*(T)$	§3.5	the 0.09% agreement is otherwise a coincidence

Standard algebraic derivations (Lemma 1, Lemma 4, Proposition 3, the ledger identity of §2.7, and the reliability product law cited in §1.6) are not repeated here; they are one-line substitutions and are stated in the body with their proofs.

Throughout, $S_0 = 0.5$, $v(S) = V_{\max} S/(K_s + S + S^2/K_i)$ with $(V_{\max}, K_s, K_i) = (10, 1, 4)$, so $v(S_0) = 3.2$; $C_0 = C(1) = 1.84255$; $R(t) = (1-\varepsilon)^{\lfloor t/P \rfloor}$ with $\varepsilon = 0.01$ and $P = 10$; $\sigma_v = 0.08$, $\sigma_m = 0.12$; $\kappa = 3$; $A_{\text{sleep}} = 2$; $L = 2$; $P_{\text{sh}} = 23$.

A.7 (A7) There is no parameter path from M1 to M2

This is the first result of the appendix because it is the one that licenses the paper's central structural claim. If a sequence of parameters existed along which the dormant law converges to the active law, then "two regimes" would be an artefact of our parameter choice, the join A7

would be a limit rather than an assumption, and the paper's methodological conclusion in §4.3 would be wrong. The proposition below says no such sequence exists.

Setting. Let λ range over the model parameters $(V_{\max}, K_s, K_i, Y, \phi_v, \phi_{\text{dorm}}, m_0, D_0)$. Write

$$f_\lambda(A,t) = Y\phi_v\, v_\lambda\big(S(t)\big)\,(1-A) - \phi_{\text{dorm}}\, m_0, \qquad g_\lambda(t) = \frac{m_0}{D_0}\big[(1+\epsilon_m)d_\lambda(t) - c_\lambda(t)\big],$$

for the right-hand sides of `Ax2` and `Ax3`. Fix one shock-and-drift realisation $S(\cdot)$ and one delivery/commitment realisation $d_\lambda(\cdot), c_\lambda(\cdot)$ as in §2.9.

Proposition A.7.1 (no nontrivial uniform limit). Suppose there is a sequence λ_n and an interval of cycles $\mathcal{I}$ such that

$$\sup_{A\in K}\ \sup_{t\in\mathcal{I}}\ \big|f_{\lambda_n}(A,t) - g_{\lambda_n}(t)\big| \ \xrightarrow[n\to\infty]{}\ 0$$

for some compact $K \subset \mathbb{R}$ with nonempty interior. Suppose also that g_{λ_n} changes sign on $\mathcal{I}$ for every n: there exist cycles $t_n^-, t_n^+ \in \mathcal{I}$ with $g_{\lambda_n}(t_n^-) < 0 < g_{\lambda_n}(t_n^+)$. Then $\sup_{t\in\mathcal{I}} |g_{\lambda_n}(t)| \to 0$.

That is: the only limits are the trivial ones, in which both vector fields vanish identically and neither system moves.

Proof. Write $a_n(t) = Y\phi_v v_{\lambda_n}(S(t)) \geq 0$ and $b_n = \phi_{\text{dorm}} m_0(\lambda_n) \geq 0$, so that $f_{\lambda_n}(A,t) = a_n(t)(1-A) - b_n$. For $A, A' \in K$ with $A \neq A'$,

$$f_{\lambda_n}(A,t) - f_{\lambda_n}(A',t) = -a_n(t)(A - A').$$

On the other hand g_{λ_n} does not depend on A, so

$$\big|a_n(t)(A-A')\big| = \big|\big(f_{\lambda_n}(A,t) - g_{\lambda_n}(t)\big) - \big(f_{\lambda_n}(A',t) - g_{\lambda_n}(t)\big)\big| \leq 2\sup_{A\in K}\big|f_{\lambda_n} - g_{\lambda_n}\big|.$$

Dividing by $|A - A'| > 0$ gives $a_n \to 0$ uniformly on $\mathcal{I}$. Hence

$$f_{\lambda_n}(A,t) = a_n(t)(1-A) - b_n \ \longrightarrow\ -b_n$$

uniformly in (A,t). Since $f_{\lambda_n} \to g_{\lambda_n}$ uniformly, the triangle inequality gives

$$\sup_{t\in\mathcal{I}}\big|g_{\lambda_n}(t) + b_n\big| \ \longrightarrow\ 0.$$

Now b_n is a constant in t, so the limit says $g_{\lambda_n}(t) = -b_n + \delta_n(t)$ with $\sup_t |\delta_n| \leq \omega_n$, where $\omega_n = \sup_{A\in K}\sup_t \big|f_{\lambda_n}(A,t) - g_{\lambda_n}(t)\big| \to 0$. Evaluate at the two sign-changing cycles: $0 < g_{\lambda_n}(t_n^+) = -b_n + \delta_n(t_n^+) \leq -b_n + \omega_n$ gives $b_n < \omega_n$, and $0 > g_{\lambda_n}(t_n^-) = -b_n + \delta_n(t_n^-) \geq -b_n - \omega_n$ is implied by it. Hence $0 \leq b_n < \omega_n \to 0$, and then

$$\sup_t |g_{\lambda_n}(t)| \leq b_n + \omega_n \longrightarrow 0.$$

That is the conclusion, not a contradiction: a sign change forces the constant term to vanish, so both fields go to zero together, the trivial limit $(0,0)$.

Both types realise the hypothesis. For Type-A, d is atomic, so an interval containing a cycle with $d = c > 0$ and a shock cycle with $d = 0$, $c > 0$ has $g = \frac{m_0}{D_0}\epsilon_m c > 0$ and $g = -\frac{m_0}{D_0}c < 0$; naming that pair explicitly, as an earlier formulation did, made the proposition look Type-A-specific. For Type-B, $d = \min(c, C)$, so $g = -\frac{m_0}{D_0}\big(c - (1+\epsilon_m)C\big)$ when $C < c$ and $+\frac{m_0}{D_0}\epsilon_m c$ otherwise; the sign change then needs one cycle with $C < c/(1+\epsilon_m)$ — in particular $C = 0$ — and one with $C \geq c$. Both occur in the reference scenario, and the proof above invokes only the signs.

Corollary A.7.2 (the fixed points cannot be matched either). Even in the trivial limit $a_n \to 0$, $b_n \to 0$, the two systems are both the identity map, and the statement "M1's fixed point converges to M2's" is vacuous. Outside that limit, M1 has the unique fixed point $A^* = 1 - b/a$ (which exists iff $a > 0$) and M2 has no fixed point whenever $g \not\equiv 0$. There is no parameter value at which the two fixed-point sets agree and are nonempty.

Corollary A.7.3 (what this licenses). The join `Ax6` is an assumption, not a derivation; the two-regime model is the correct description and not a failure to find the right parameterisation; and any claim of the form "the metabolic law implies grounding is necessary" is unsupported, because the extinction at cycle 107 is a property of g (via $d = 0$), not of f.

Remark A.7.4 (why the proof is not an artefact of the shock schedule). Only two cycles with opposite signs of $(1 + \epsilon_m)d - c$ are required. In the base run, $d = c$ holds on 4550 agent-cycles and $d = 0$ on 708, so $\mathcal{I}$ may be taken as any window containing one of each. The argument therefore applies to every run in this paper, and to every run with at least one `PASS` and one `INVALID` or `FAIL`.

Remark A.7.5 (what is *not* claimed). The proposition does not say that M1 and M2 are unrelated. They share the substrate variable, the death event, and the switch; `Ax6` couples them. What is claimed is narrower and sufficient: no smooth deformation of the parameters identifies one with the other, so the coupling is a modelling decision that must be declared.

A.1 (A1) Closed form for the wake-gate crossing

Lemma A.1.1 (monotone inversion). v is C^∞ and strictly increasing on $(0, \sqrt{K_s K_i}) = (0, 2)$ with $v(0^+) = 0$ and $v(2) = V_{\max}/2 = 5$. For every $\phi \in (0, 1)$ there is a unique $S_\phi \in (0, S_0)$ with $v(S_\phi) = \phi\, v(S_0)$, because $\phi v(S_0) < v(S_0)$ and v is strictly increasing there.

Proof. $v'(S) = V_{\max}(K_s - S^2/K_i)/(K_s + S + S^2/K_i)^2 > 0$ on $(0, \sqrt{K_s K_i}) = (0, 2)$; the intermediate value theorem gives existence and strict monotonicity gives uniqueness. □

Proposition A.1.2 (peak gate). Let the gate be $W_i(t) \geq \phi W_i(1)$ with $W_i(t) = v(S_0 R(t))\, v_i$ and $W_i(1) = v(S_0)\, v_i$. Since v_i cancels, the gate is agent-independent. It holds iff $R(t) \geq S_\phi/S_0$, and fails from

$$t_g = P \left\lceil \frac{\ln(S_\phi/S_0)}{\ln(1-\varepsilon)} \right\rceil .$$

Proof. $v(S_0 R) \geq \phi v(S_0) \iff S_0 R \geq S_\phi$ by Lemma A.1.1 applied on the increasing branch. With $R = (1-\varepsilon)^{\lfloor t/P \rfloor}$ and $S_\phi < S_0$, the inequality $\lfloor t/P \rfloor \leq K$ with $K = \ln(S_\phi/S_0)/\ln(1-\varepsilon) > 0$ holds iff $t < P(K+1)$, i.e. fails first at $t = P\lceil K \rceil$. □

Numerical check. $\phi = 0.85$: $S_{0.85} = 0.3876638$, $S_\phi/S_0 = 0.7753276$, $K = 25.3195$, $t_g = 260$. This is the first cycle on which the gate is violated, and the value is not fitted.

Proposition A.1.3 (mean gate). With $C(r) = \frac{1}{n}\sum_{j=0}^{n-1} v\big(S_0 r(1-\delta_s)^j\big)$, define r_ϕ as the unique root of $C(r_\phi) = \phi C(1)$. Then $t_g = P\lceil \ln r_\phi / \ln(1-\varepsilon) \rceil$.

Proof. C is a positive combination of strictly increasing functions of r, hence strictly increasing, and $C(0^+) = 0 < \phi C(1) < C(1)$. Apply the same floor argument as in Proposition A.1.2. □

Numerical check. $r_{0.85} = 0.8084427$, $K = 21.1580$, $t_g = 220$.

Proposition A.1.4 (the legacy reading, and why it is not a reparameterisation). Let the legacy gate be $W_i(t) \geq \phi C(1) v_i$, i.e. a peak compared with a mean. Then $t_g = 970$. The three readings are not related by a monotone reparameterisation of ϕ: the two functionals $r \mapsto v(S_0 r)$ and $r \mapsto C(r)$ are not proportional, since

$$\frac{v(S_0 r)}{C(r)} = \frac{3.2}{1.84255} = 1.7367 \text{ at } r = 1, \qquad \frac{v(S_0 r)}{C(r)} = \frac{2.7778}{1.5533} = 1.7883 \text{ at } r = 0.8.$$

Hence no single scalar ϕ can place the two thresholds at the same r, and the legacy number cannot be obtained from the correct one by rescaling.

Numerical check. $S_{0.85} \cdot (C_0/v_0)$-threshold: $v(S_0 r) = 0.85 C_0 = 1.56617$ gives $r = 0.3769579$, $K = 96.902$, $t_g = 970$.

Table A.1 collects the three readings side by side, so that the columns can be checked against the propositions above.

Table A.1. The three gate readings, decomposed.

mode	signal	level	r at crossing	K	t_g	t_{dorm}	t_{strand}
`peak`	$v(S_0 r)$	$0.85\, v(S_0)$	0.775328	25.32	260	277	279
`mean`	$C(r)$	$0.85\, C(1)$	0.808443	21.16	220	231	233
`legacy`	$v(S_0 r)$	$0.85\, C(1)$	0.376958	96.90	970	981	983

The last two columns are computed in §A.2 and agree with the simulation in all three rows.

A.2 (A2) Stranding latency: from the boundary to the event

Proposition A.1.2 locates the *boundary*: the first cycle after which the gate can never be met again. It does not locate the *event*, because the predicate ST (D5) is evaluated only inside a dormancy episode. Proposition A.2.2 closes that gap, and Proposition A.2.3 identifies the entry time t_{dorm} on which it depends.

Lemma A.2.1 (accumulation). Let $s(t)$ be the counter of consecutive dormant cycles on which the gate is violated, reset to 0 on any dormant cycle on which it is met and untouched on active cycles. Then

$$t_{\text{strand}} = \min\{t : \ s(t) \geq \kappa\},$$

and s can only increase on a dormant cycle after t_g.

Proof. The implementation increments `strand_run` in `_dormant_cycle` only, and resets it in the two branches where the gate is met. Since the gate is monotone (Proposition A.1.1 and R nonincreasing), no reset can occur after t_g. □

Proposition A.2.2 (latency). If the agent is dormant at t_g, or if t_{dorm} denotes the first dormancy entry at or after t_g, then

$$t_{\text{strand}} = t_{\text{dorm}} + (\kappa - 1).$$

Proof. By Lemma A.2.1 and monotonicity of R, s increments on every dormant cycle from t_{dorm} onward, so $s(t_{\text{dorm}} + j) = j + 1$ and the threshold κ is first reached at $j = \kappa - 1$. □

The remaining quantity is t_{dorm}. Entry into dormancy is governed by the failure streak (the charge stays well above θ_{dorm} in every run here, so the low-charge clause never fires), and a failure occurs exactly on cycles where the claim exceeds the capacity.

Proposition A.2.3 (entry time). Define

$$t_0 = \min\Big\{\tau \geq t_g : \ c_i(\tau) > C_i(\tau)\Big\}$$

as the first cycle at or after the boundary on which the agent's claim exceeds the capacity. Assume $A_{\text{sleep}} \leq L$, so that consecutive failures can accumulate inside a single shock episode. Then

$$t_{\text{dorm}} = t_0 + \big(A_{\text{sleep}} - 1\big).$$

Proof. The streak increments on each failing cycle and resets on each success. By Proposition A.1.2 and the monotonicity of R, once $c_i > C_i$ holds at t_0 it holds for all $\tau \geq t_0$: capacity is nonincreasing and the AIMD loop can only shrink the claim, by 5% per failure. Hence the streak reaches A_{sleep} at $t_0 + A_{\text{sleep}} - 1$. □

Remark A.2.4 (why t_0 has no closed form, and what replaces it). The claim $c_i(\tau)$ is set by the AIMD loop, not by the environment, so t_0 cannot be written as a function of the schedules alone. What can be said is quantitative and sufficient: the loop keeps c_i/C_i within a few parts in a thousand of unity, so t_0 is the first cycle after t_g at which capacity steps down by more than that residual. Two step sizes compete — a 1% drift step every 10 cycles and a 15% shock every 23 cycles — and which one fires first depends only on where t_g falls in the two schedules. Measuring the residual once per run (it is 0.8%–0.9% in all three runs below) fixes t_0 completely.

Verification. Table A.2 evaluates this in all three gate readings.

Table A.2. Verified stranding latency.

mode	t_g	next drift step	next shock	residual gap	t_0	t_{dorm}	t_{strand}	observed
`peak`	260	270 (0.80%)	276 (15%)	$> 0.80\%$	276	277	279	279
`mean`	220	230 (1%)	230 (15%)	any	230	231	233	233
`legacy`	970	980 (0.89%)	989 (15%)	$< 0.89\%$	980	981	983	983

In the `mean` row the drift step and the shock coincide, because 230 is a multiple of both 10 and 23; this is the only coincidence of the two schedules below 300 and it is why the `mean` value is the least sensitive of the three to the residual. In the `peak` row the drift step at 270 is smaller than the residual and the shock at 276 is not, so the shock is the trigger. In the `legacy` row the roles are reversed: the drift step at 980 arrives first and is already larger than the residual, so the shock at 989 is never reached.

Corollary A.2.4 (why the 50% figures are all-or-nothing). The gate is agent-independent, because v_i and $m_{0,i}$ cancel from both sides of $W_i(t) \geq \phi W_i(1)$. Hence every agent's gate fails on the same cycle, every agent's stranding event is governed by the same t_g, and the stranded count is either 0 or the full population. This is why Table 4 reports 50 and not an intermediate number for $\theta \leq 0.90$, and it is a consequence of the gate's form, not of the population's homogeneity.

Remark A.2.5 (what A2 does not cover). The proposition uses $A_{\text{sleep}} \leq L$. With $L < A_{\text{sleep}}$ entry would require two episodes and t_{dorm} would gain one shock period; the experiments in this paper all satisfy $A_{\text{sleep}} = L = 2$, so the case does not arise. It is recorded here because the relation is fragile under a change of L, and §3.7 reports that L is exactly the parameter that moves the extinction.

A.5 (A5) The order-statistic law of the boundary

The boundary panel reports the fraction of agents that deadlock at each θ_{wake}. Proposition A.5.2 gives that fraction in closed form, and Corollary A.5.3 shows that its 50% point is *exactly* the terminal anchor $\theta^*(T)$ of Corollary 2. The agreement reported in §3.5 is therefore an identity, not a fit.

Lemma A.5.1 (individual margin). Write $v_i = \exp \xi_i$, $m_{0,i} = m_0 \exp \zeta_i$ with $\xi_i \sim \mathcal{N}(0, \sigma_v^2)$, $\zeta_i \sim \mathcal{N}(0, \sigma_m^2)$ independent. Write

$$\eta_i = \frac{v_i/v_0}{m_{0,i}/m_0} = e^{\xi_i - \zeta_i}, \qquad \rho(t) = \frac{v(S_0 R(t))}{v(S_0)}\,\mu(t),$$

where $\mu(t) \in \{\sigma_s, 1\}$ is the shock multiplier. Then

$$\chi_i(t) = \chi_0\,\rho(t)\,\eta_i, \qquad \eta_i \sim \mathrm{LN}(0, s^2), \quad s^2 = \sigma_v^2 + \sigma_m^2.$$

Proof. By D8, $\chi_i = Y\phi_v W_i/(m_{0,i}\phi_{\mathrm{dorm}})$ with $W_i = v(S_0R)\mu\, v_i$. Substituting and collecting gives the stated product; the log of a ratio of independent lognormals is normal with variance $\sigma_v^2 + \sigma_m^2$. □

Proposition A.5.2 (deadlock probability). Let $\rho_* = \min\{\rho(t) : t \le T,\ t \text{ shock-free}\}$, and suppose every agent that can satisfy the predicate enters dormancy at or after the first cycle at which it does. Then

$$\Pr[\text{DL by } T] = F_\eta\left(\frac{\chi_{\min}(\theta)}{\chi_0\,\rho_*}\right) = \Phi\left(\frac{1}{s}\ln\frac{\chi_{\min}(\theta)}{\chi_0\,\rho_*}\right), \qquad \chi_{\min}(\theta) = \frac{1}{1-\theta}.$$

Proof. By Theorem 1 and Lemma A.5.1, agent i deadlocks at time t iff $\eta_i < \chi_{\min}/(\chi_0\rho(t))$. The right-hand side is increasing in t because ρ is nonincreasing (v increasing in S, R nonincreasing, $\mu = 1$ on shock-free cycles), so the union over $t \le T$ is the event $\eta_i < \chi_{\min}/(\chi_0\rho_*)$. Taking the lognormal CDF gives the formula. □

Corollary A.5.3 (the median is the anchor). Setting the argument to 1:

$$\Pr[\text{DL}] = \tfrac{1}{2} \iff \chi_{\min} = \chi_0\rho_* \iff \theta = 1 - \frac{1}{\chi_0\rho_*} = \theta^*(T).$$

Because $\eta \sim \mathrm{LN}(0, s^2)$ has median 1, the midpoint of the boundary curve equals the terminal anchor of Corollary 2 exactly, for every s, every σ_s and every population size. The value of s moves the *width* of the transition, not its centre.

Numerical evaluation. At $T = 300$, $R = 0.7397$, $v(S_0R)/v(S_0) = 0.82322$, so $\chi_0\rho_* = 17.7778 \times 0.82322 = 14.6343$ and $\theta^*(300) = 1 - 1/14.6343 = 0.93167$. The ratio 0.82322 is the uptake map applied to R, not R itself; §3.5 states what this presupposes about the drift schedule. With $s = 0.14422$:

Table A.3. Predicted vs measured deadlock counts (50 Type-B agents).

θ_{wake}	$\chi_{\min}$	measured	pop. p	pop. count	seed p	seed count
0.90	10.000	0	0.0041	0.21	0.0007	0.03
0.91	11.111	2	0.0281	1.40	0.0086	0.43
0.92	12.500	4	0.1372	6.86	0.0726	3.63
0.93	14.286	19	0.4336	21.68	0.3427	17.14
0.94	16.667	43	0.8164	40.82	0.7907	39.53
0.95	20.000	50	0.9848	49.24	0.9876	49.38

The "pop." columns use $s = 0.14422$ and median 1, i.e. the unconditional law; the "seed" columns replace both by the realised moments of $\{\eta_i\}$ for seed 7, which are median 1.0277 and log-sd 0.1270. Neither column is fitted per row: each is a single substitution. The unconditional law is exact at 0.95 and 0.90 and over-predicts at 0.92–0.93 by 3–7 counts; the seed-conditional law tracks all six rows within 2 counts, and its largest error is at 0.94 (39.5 predicted, 43 observed). The residual at 0.91 (0.4 predicted, 2 observed) is a Poisson event.

The correct reading is that the *law* is exact and the *sweep* is a 50-agent sample. The unconditional form makes a statement about an ensemble of populations; the conditional form makes the same statement about this one, and it is the one that should be compared with a single run.

The measured midpoint. Interpolating the measured fractions between 0.93 (0.38) and 0.94 (0.86) gives 0.9325. The theoretical value is 0.9317 and the sample-conditional value, using the realised median 1.0277, is

$$1 - \frac{1}{14.6343 \times 1.0277} = 0.93351.$$

The measured value lies between the two, and the residual is attributable to the two-point interpolation rather than to the model. The 0.09% agreement quoted in §3.5 is therefore the infinite-population statement; the correct statement for this seed is that the seed's own median displaces the midpoint by $\ln(1.0277)/\chi_0\rho_* \approx 0.0018$ in θ.

Corollary A.5.4 (width, not location, is a free parameter). The transition width is set by $s = \sqrt{\sigma_v^2 + \sigma_m^2}$, i.e. by the dispersion of $\log(v_i/m_{0,i})$. With the shipped $\sigma_v = 0.08, \sigma_m = 0.12$, $s = 0.14422$ and the 10%–90% width in θ is

$$\Delta\theta_{10\text{–}90} = \theta^*\big|_{z=+1.2816} - \theta^*\big|_{z=-1.2816} = 0.94321 - 0.91779 = 0.0254,$$

evaluated at $T = 300$: $\theta^*(z) = 1 - 1/(\chi_0\rho_* e^{zs})$, so the endpoints are 0.9178 and 0.9432. A wider dispersion widens this interval with $\partial\Delta\theta_{10\text{–}90}/\partial s > 0$ but leaves its midpoint at θ^* exactly. A population with $\sigma_v = \sigma_m = 0$ has a step at θ^*; a heterogeneous population has a smooth ramp whose centre is still θ^*. This is the formal content of the statement in §3.5 that the transition is distributional and not a phase transition.

Remark A.5.5 (the one assumption in A5). The union over t in the proof requires that every agent which *can* satisfy $\chi_i < \chi_{\min}$ at some t does in fact enter dormancy. For Type-B this holds because the gate fails at t_g (Proposition A.1.2), the resulting failures trigger entry by Proposition A.2.3, and entry is permanent at the level of the gate. For Type-A it fails by construction (I4), which is why Table 4 reports no Type-A deadlocks: the zero is definitional and not evidence about the boundary.

A.28 (A28) Pre-shock saturation, and the dependency of Corollary 4

Corollary 4 in §2.6 computes the death cycle from a saturation reference t_s at which $A_i = 1$ for every Type-A agent, and takes $t_s = 91$. That step presumes the agents have saturated before the first shock. The presumption is stated here as an explicit assumption because it is a premise of a quantitative formula and not a theorem of the axioms.

Assumption A28. Every Type-A agent attains $A = 1$ under the projection before the first shock cycle.

Proposition A.28.1 (A28 holds in the reference scenario). Under `M2` with $d = c = D_{0,i}$ (the `PASS` branch),

$$A_i(t+1) = A_i(t) + \frac{m_{0,i}}{D_{0,i}}\,\epsilon_m\, D_{0,i} = A_i(t) + \epsilon_m\, m_{0,i},$$

so with $A_i(0) = 0.85$ the saturation time is $t_{\text{sat},i} = \lceil (1 - A_0)/(\epsilon_m m_{0,i}) \rceil = \lceil 1/m_{0,i} \rceil$, using $\epsilon_m = 0.15$ and $A_0 = 0.85$ so that $(1 - A_0)/\epsilon_m = 1$. With $m_{0,i} \in [0.1162, 0.1829]$ this gives $t_{\text{sat},i} \in [6, 9]$, and the first shock is at cycle 23. Hence A28 holds with a margin of 14 cycles, and the last saturated cycle is 91 only because the *last* re-saturation after the shock of 92–93 is at 91.

Dependency chain. A28 is used by Corollary 4; Corollary 4 is used by §2.6 and §3.2; nothing else depends on it. A28 is therefore appended to the inventory and is *not* added to the eleven load-bearing assumptions, because no theorem fails if it fails — only the closed-form multiplicities $11, 23, 13, 3$ in §3.2 become a numerical observation instead of a derivation.

Remark A.28.2 (when A28 breaks). The margin is $(1 - A_0)/(\epsilon_m m_{0,\min})$ versus the first shock at $P_{\text{sh}} = 23$. A28 fails if ϵ_m is reduced by a factor above $23/9 \approx 2.6$ (to $\epsilon_m \lesssim 0.058$), or if A_0 is lowered substantially, or if the first shock is moved earlier than cycle 9. In any of those cases Corollary 4 still holds at the level of a single agent, but the deficit δ becomes agent-dependent and the closed form must be applied per agent rather than to the population.

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